

# WEYL-CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL-RIESZ TRANSFORM COMMUTATORS

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ABSTRACT.

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## 1. INTRODUCTION

**Theorem 1.1.** *Let  $R$  be a normalised root system in  $\mathbb{R}^N$ ,  $N \geq 2$ , and let  $G$  be the corresponding finite reflection group. Let  $\kappa : R \rightarrow \mathbb{R}_+$  be  $G$ -invariant multiplicity function. Let  $R_{\text{Dunkl},j}$  be the  $j$ -th Dunkl-Riesz transform corresponding to  $R$  and  $k$ . Let  $w$  be the Dunkl weight corresponding to  $R$  and  $k$ . Let  $f \in L_1^{\text{loc}}(\mathbb{R}^N)$ .*

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- (i) If  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ , then  $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ . Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_\kappa \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

- (ii) If  $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ , then  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ . Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \geq c_\kappa^{-1} \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

## 2. PRELIMINARIES

**2.1. Harmonic analysis in the Dunkl setting.** Let  $\mathbb{R}^N$  be the Euclidean space equipped with the usual scalar product  $\langle \cdot, \cdot \rangle$  and the induced norm  $\|\cdot\|$ . For a nonzero vector  $v \in \mathbb{R}^N$ , the reflection  $\sigma_v$  with respect to the hyperplane orthogonal to a normalised vector  $v$  is defined by

$$\sigma_v x := x - 2\langle x, v \rangle v.$$

A finite set  $R \subset \mathbb{R}^N \setminus \{0\}$  is called a root system if  $\sigma_v(R) = R$  for every  $v \in R$ . A root system  $R$  is called a normalised reduced root system if  $\|v\|^2 = 1$  for every  $v \in R$ . The finite group  $G$  generated by the reflections  $\{\sigma_v\}_{v \in R}$  is called the Weyl group (reflection group) of the root system  $R$ .

Let  $\kappa : R \rightarrow [0, \infty)$  be a multiplicity function, which is a  $G$ -invariant function fixed throughout the paper.

Define  $d(x, y) = \min_{g \in G} |x - gy|$ . Recall from [8, Chapter VII, proof of Theorem 2.12] that  $d(x, y) = |x - y|$  iff  $x$  and  $y$  belong to the same chamber.

Define the Dunkl weight on  $\mathbb{R}^N$  by

$$w(x) = \prod_{v \in R} |\langle x, v \rangle|^{\kappa(v)}, \quad x \in \mathbb{R}^N.$$

Dunkl measure  $w(x)dx$  is also denoted by  $w$ .

Let  $B(x, r) := \{y \in \mathbb{R}^N : \|x - y\| < r\}$  be the Euclidean ball centred at  $x$  with radius  $r > 0$ . Let  $\mathbf{N} := N + \sum_{v \in R} \kappa(v)$ . Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \quad \text{for } x \in \mathbb{R}^N, \quad t, r > 0.$$

Furthermore,

$$(1) \quad w(B(x, r)) \sim r^{\mathbf{N}} \prod_{v \in R} (|\langle x, v \rangle| + r)^{\kappa(v)},$$

where the implicit constant only depends on  $N, R$  and  $\kappa$ . It follows from (1) that there exists a constant  $C \geq 1$  such that for every  $x \in \mathbb{R}^N$ ,

$$(2) \quad C^{-1} \left( \frac{r_2}{r_1} \right)^{\mathbf{N}} \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left( \frac{r_2}{r_1} \right)^{\mathbf{N}}, \quad \text{for } 0 < r_1 < r_2,$$

so the numbers  $\mathbf{N}$  and  $N$  are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators  $T_\xi$  are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{\kappa(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where  $\partial_\xi f$  denotes the directional derivative of  $f$  in the direction  $\xi$ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} = - \sum_{j=1}^N T_{e_j}^2.$$

In this article, we consider the Dunkl-Riesz transforms defined by  $R_{\text{Dunkl},j} = T_{e_j} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}$ ,  $j = 1, 2, \dots, N$ .

Operators  $U_g : f \rightarrow f \circ g^{-1}$ ,  $g \in G$ , are unitary both in  $L_2(\mathbb{R}^N)$  and in  $L_2(\mathbb{R}^N, w(x)dx)$ . Obviously,  $g \rightarrow U_g$  is a unitary representation of the group  $G$ . To lighten the notations, we write  $U_v$  instead of  $U_{\sigma_v}$  for  $v \in R$ .

## 2.2. Weyl chamber as simplicial cone.

**Definition 2.1.** A connected component of the set

$$\{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}$$

is called an open Weyl chamber. Its closure is called a Weyl chamber.

The following assertion can be found in e.g. Theorem 2 on p.166 in [1].

**Theorem 2.2.** Let  $\Gamma$  be a chamber. There exists a linear isomorphism  $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$  such that  $\Phi(\Gamma) = R_+^{N_1} \times \mathbb{R}^{N_2}$  (with  $N_1 + N_2 = N$ ).

## 2.3. System of dyadic cubes.

**Notation 2.3.** Let

$$\mathbb{D} = \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k$$

be the standard system of dyadic cubes over  $\mathbb{R}^N$ . Here each dyadic cube in  $\mathbb{D}_k$  is of the shape

$$\frac{m}{2^k} + [0, 2^{-k})^N, \quad m \in \mathbb{Z}^N.$$

For every dyadic cube  $Q \in \mathbb{D}$ , we denote by  $c_Q$  and  $\ell(Q)$  its center and side-length, respectively.

**2.4. Nearly weakly orthonormal sequences.** Rochberg and Semmes [13] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [10, 12, 3, 4, 11, 13, 15, 14]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [10, Section 13] for a complete proof).

Definition 2.4 and Lemma 2.5 as stated below are my understanding of Hytonen's Corollary 8.2 and Lemma 8.7. Please check whether this understanding is the correct one!

**Definition 2.4.** Let  $\mathbf{w} \in A_2(\mathbb{R}^N)$ . Let  $c > 0$  and  $2 < r \leq \infty$ . A sequence of functions  $\{e_Q\}_{Q \in \mathbb{D}}$  is called  $(r, c)$ -NWO if

- (i)  $\text{supp}(e_Q) \subseteq cQ$  for every  $Q \in \mathbb{D}$ ;
- (ii)  $\|e_Q\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} \leq \mathbf{w}(Q)^{\frac{1}{r} - \frac{1}{2}}$  for every  $Q \in \mathbb{D}$ ;

**Lemma 2.5.** *Let  $\mathbf{w} \in A_2(\mathbb{R}^N)$ . Let  $0 < p < \infty$ . Suppose that  $A$  is an integral operator on  $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$  with integral kernel*

$$(x, y) \rightarrow \sum_{Q \in \mathbb{D}} \lambda_Q e_Q(x) f_Q(y), \quad x, y \in \mathbb{R}^N,$$

*where  $\{e_Q\}_{Q \in \mathbb{D}}$  and  $\{f_Q\}_{Q \in \mathbb{D}}$  are  $(r_1, c_1)$ -NWO and  $(r_2, c_2)$ -NWO sequences, respectively. We have*

$$\begin{aligned} \|A\|_{\mathcal{L}_{p,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))} &\leq C_{w,r_1,c_1,r_2,c_2,p} \|\{\lambda_Q\}_{Q \in \mathbb{D}}\|_{p,\infty}, \\ \text{dist}_{\mathcal{L}_{p,\infty}(L_2(\Gamma, \mathbf{w}(x)dx))} \left( A, (\mathcal{L}_{p,\infty})_0(L_2(\mathbb{R}^N, \mathbf{w}(x)dx)) \right) &\leq \\ &\leq C_{\mathbf{w},r_1,c_1,r_2,c_2,p} \text{dist}_{l_{p,\infty}(\mathbb{D})} \left( \{\lambda_Q\}_{Q \in \mathbb{D}}, (l_{p,\infty})_0(\mathbb{D}) \right). \end{aligned}$$

*Proof.* The assertion follows from Corollary 8.2 and Lemma 8.7 in [10].  $\square$

**Lemma 2.6.** *Let  $w \in A_2(\mathbb{R}^N)$ . Let  $1 < p < \infty$ . Suppose that  $A$  is an integral operator on  $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$  with integral kernel*

$$(x, y) \rightarrow \sum_{Q \in \mathbb{D}} \lambda_Q e_Q(x) f_Q(y), \quad x, y \in \mathbb{R}^N,$$

*where  $\{e_Q\}_{Q \in \mathbb{D}}$  and  $\{f_Q\}_{Q \in \mathbb{D}}$  are  $(r_1, c_1)$ -NWO and  $(r_2, c_2)$ -NWO sequences, respectively. We have*

$$\left\| \left\{ \langle A e_Q, f_Q \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{p,q}(\mathbb{D})} \leq c_{\mathbf{w},r_1,c_1,r_2,c_2,p,q} \|A\|_{\mathcal{L}_{p,q}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))}.$$

*Proof.* The assertion follows from Corollary 8.2 and Lemma 8.7 in [10].  $\square$

**2.5. Principal symbol and crossed products.** Let  $\Pi$  be the  $C^*$ -subalgebra in  $B(L_2(\mathbb{R}^N))$  generated by  $M_f$ ,  $f \in (C_0 + \mathbb{C})(\mathbb{R}^N)$  and  $h(\frac{\nabla}{\sqrt{\delta}})$ ,  $h \in C(\mathbb{S}^{N-1})$ . Note that every compact operator is contained in  $\Pi$ . There exists a surjective  $*$ -homomorphism

$$\text{sym} : \Pi \rightarrow (C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})$$

such that  $\ker(\text{sym})$  consists exactly of all compact operators.

We have a natural action  $\alpha$  of  $G$  on  $B(L_2(\mathbb{R}^N))$  and, therefore, on  $\Pi$  — the  $C^*$ -algebra of operators admitting principal symbol. Namely,

$$\alpha(g) : T \rightarrow U_g T U_g^{-1}, \quad T \in B(L_2(\mathbb{R}^N)), \quad g \in G.$$

Define the action  $\beta$  of  $G$  on  $(C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})$  by setting

$$(\beta(g))(f \otimes h) = (f \circ g^{-1}) \otimes (h \circ g^{-1}).$$

The same action extends to von Neumann algebra  $L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1})$ . We have an equivariance relation

$$\text{sym}(\alpha(g)(T)) = (\beta(g))(\text{sym}(T)).$$

Clearly,  $\text{sym}$  is surjective,  $\ker(\text{sym})$  is  $\alpha$ -invariant. Hence, there exists a natural  $*$ -homomorphism

$$\pi \rtimes G : \Pi \rtimes_\alpha G \rightarrow ((C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})) \rtimes_\beta G.$$

We have an inclusion

$$((C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})) \rtimes_\beta G \subset (L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1})) \rtimes_\beta G.$$

Hence,

$$\pi \rtimes G : \Pi \rtimes_{\alpha} G \rightarrow (L_{\infty}(\mathbb{R}^N) \bar{\otimes} L_{\infty}(\mathbb{S}^{N-1})) \rtimes_{\beta} G.$$

Next, the standard trace  $\int_{\mathbb{R}^N} \otimes \int_{\mathbb{S}^{N-1}}$  on  $L_{\infty}(\mathbb{R}^N) \bar{\otimes} L_{\infty}(\mathbb{S}^{N-1})$  is  $\beta$ -invariant. Hence, there exists a natural trace

$$\left( \int_{\mathbb{R}^N} \otimes \int_{\mathbb{S}^{N-1}} \right) \rtimes G$$

on  $(L_{\infty}(\mathbb{R}^N) \bar{\otimes} L_{\infty}(\mathbb{S}^{N-1})) \rtimes_{\beta} G$ .

### 3. CHAMBER IS A DOUBLING METRIC MEASURE SPACE

**Theorem 3.1.** *Let  $\Gamma$  be a chamber. For all  $x \in \Gamma$  and for all  $r > 0$ , we have*

$$w(B(x, r) \cap \Gamma) \geq c_{\kappa} w(B(x, r)).$$

*In particular,  $(\Gamma, |\cdot|, w(x)dx)$  is a doubling metric measure space.*

**Notation 3.2.** *Let  $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ . Set*

$$w_{\theta}(x) = \prod_{A \subset \{1, \dots, N_1\}} \left( \sum_{k \in A} x_k \right)^{\theta_A}, \quad x \in \mathbb{R}_+^{N_1}.$$

**Lemma 3.3.** *For every  $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$  and for every  $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ , we have*

$$\int_{B(x, r)} w_{\theta}(y_1, \dots, y_{N_1}) dy \leq c_{N, \theta}^{(1)} r^N w_{\theta}((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}).$$

*Proof.* Obviously,

$$B(x, r) \subset \left( \times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r).$$

Thus,

$$\begin{aligned} & \int_{B(x, r)} w_{\theta}(y_1, \dots, y_{N_1}) dy \leq \\ & \leq \int_{\left( \times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r)} w_{\theta}(y_1, \dots, y_{N_1}) dy = \\ & = \int_{\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r]} w_{\theta}(u) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \\ & \leq \int_{\times_{k=1}^{N_1} [0, x_k + r]} w_{\theta}((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} = \\ & = w_{\theta}((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times \prod_{k=1}^{N_1} (x_k + r - (x_k - r)_+) \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \\ & \leq w_{\theta}((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times (2r)^{N_1} \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2}. \end{aligned}$$

□

**Lemma 3.4.** *For every  $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$  and for every  $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ , we have*

$$\int_{B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_{\theta}(y_1, \dots, y_{N_1}) dy \geq c_{N, \theta}^{(2)} r^N w_{\theta}((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}).$$

*Proof.* Obviously,

$$x + [\frac{r}{2N}, \frac{r}{N}]^N \subset B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}).$$

It follows that

$$\begin{aligned} & \int_{B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_\theta(y_1, \dots, y_{N_1}) dy \geq \\ & \geq \int_{x + [\frac{r}{2N}, \frac{r}{N}]^N} w_\theta(y_1, \dots, y_{N_1}) dy = \\ & = \int_{(x_1, \dots, x_{N_1}) + [\frac{r}{2N}, \frac{r}{N}]^{N_1}} w_\theta(u) du \times (\frac{r}{2N})^{N_2} \geq \\ & \geq \int_{(x_1, \dots, x_{N_1}) + [\frac{r}{2N}, \frac{r}{N}]^{N_1}} w_\theta((x_1, \dots, x_{N_1}) + \frac{r}{2N} \mathbf{1}_{N_1}) du \times (\frac{r}{2N})^{N_2} = \\ & = w_\theta((x_1, \dots, x_{N_1}) + \frac{r}{2N} \mathbf{1}_{N_1}) \times (\frac{r}{2N})^N \geq \\ & \geq w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times (\frac{r}{2N})^N \times (2N)^{-\sum_{A \subset \{1, \dots, N_1\}} \theta_A}. \end{aligned}$$

□

**Lemma 3.5.** *There exists  $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$  such that*

$$c_\kappa^{-1} w_\theta(x_1, \dots, x_{N_1}) \leq w(\Phi^{-1}(x)) \leq c_\kappa w_\theta(x_1, \dots, x_{N_1}), \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

*Proof.* For every  $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$  and for every  $v \in R$ , we have

$$\langle \Phi^{-1}(x), v \rangle = \langle x, (\Phi^{-1})^*(v) \rangle.$$

The latter expression has the same sign (positive or negative) on the whole cone  $\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ . Hence,

$$((\Phi^{-1})^*(v))_k = \begin{cases} \text{something positive,} & 1 \leq k \leq N_1 \\ 0, & k > N_1 \end{cases}$$

Let  $A_v$  be the set of those  $1 \leq k \leq N_1$  with  $((\Phi^{-1})^*(v))_l$  being strictly positive. We, obviously, have

$$c_{\Phi, v}^{-1} \sum_{k \in A_v} x_k \leq |\langle \Phi^{-1}(x), v \rangle| \leq c_{\Phi, v} \sum_{k \in A_v} x_k, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

Thus,

$$\prod_{v \in R} c_{\Phi, v}^{-\kappa(v)} \left( \sum_{k \in A_v} x_k \right)^{\kappa(v)} \leq w(x) \leq \prod_{v \in R} c_{\Phi, v}^{\kappa(v)} \left( \sum_{k \in A_v} x_k \right)^{\kappa(v)}, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

This completes the proof. □

*Proof of Theorem 3.1.* The first assertion follows by combining Lemmas 3.3, 3.4 and 3.5. The second assertion is an obvious consequence of the first one. □

## 4. ROCHBERG-SEMMES BASED ESTIMATE

**Theorem 4.1.** *Let  $N \geq 2$ ,  $1 \leq N_1 \leq N$  and  $N_2 = N - N_1$ . Let  $\Omega = \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ . Let  $\mathbf{w} \in A_2(\Omega)$  be such that*

$$\sup_{x \in Q} \mathbf{w}(x) \leq c_w \frac{\mathbf{w}(Q)}{m(Q)}, \quad Q \in \mathbb{D}, \quad Q \subset \Omega.$$

*Suppose that  $K : \Omega \times \Omega \rightarrow \mathbb{C}$  satisfies*

$$|\partial_x^\alpha \partial_y^\beta K|(x, y) \leq c_{K, \alpha, \beta} \frac{|x - y|^{1 - |\alpha|_1 - |\beta|_1}}{\mathbf{w}(B(x, |x - y|))}, \quad x, y \in \Omega, \quad \alpha, \beta \in \mathbb{Z}_+^N.$$

*Let  $V$  be an integral operator on  $L_2(\Omega, \mathbf{w}(x)dx)$  with integral kernel  $K$ . Suppose  $N > 2s$ . For every  $f \in C_c^\infty(\mathbb{R}^N)$ , we have*

$$\|M_f V\|_{\mathcal{L}_{N, \infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq c_{K, \mathbf{w}} \|f\|_{L_N(\Omega)}.$$

**Lemma 4.2.** *There exists a smooth function  $\psi$  supported on  $[-1, 1]^N \setminus (-\frac{1}{4}, \frac{1}{4})^N$  such that*

$$1 = \sum_{Q \in \mathbb{D}} \chi_Q(x) \chi_{3Q}(y) \psi\left(\frac{x - y}{\ell(Q)}\right).$$

*Proof.* Choose  $\rho \in C_c^\infty(\mathbb{R}^N)$  such that  $\rho = 1$  on  $(-\frac{1}{2}, \frac{1}{2})^N$  and  $\rho = 0$  outside  $(-1, 1)^N$ . Set  $\psi(z) = \rho(z) - \rho(2z)$ . Clearly,

$$\text{supp}(\psi) \subset \left\{ z \in \mathbb{R}^N : \frac{1}{4} \leq |z|_\infty \leq 1 \right\}.$$

For every  $z \neq 0$ , we have

$$(3) \quad \sum_{j \in \mathbb{Z}} \psi(2^j z) = 1.$$

For every  $x$ , there exists a unique  $Q \in \mathbb{D}_j$  containing  $x$ . For this  $Q$ , we have  $\ell(Q) = 2^{-j}$ . We have

$$\begin{aligned} 1 &= \sum_{j \in \mathbb{Z}} \psi(2^j(x - y)) = \sum_{j \in \mathbb{Z}} \left( \sum_{Q \in \mathbb{D}_j} \chi_Q(x) \right) \psi(2^j(x - y)) = \\ &= \sum_{j \in \mathbb{Z}} \sum_{Q \in \mathbb{D}_j} \chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right) = \sum_{Q \in \mathbb{D}} \chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right). \end{aligned}$$

If  $x \in Q$  and  $\psi(\frac{x-y}{\ell(Q)}) \neq 0$ , then

$$x \in Q, \quad |x - y|_\infty \leq \ell(Q).$$

We obviously have  $|x - c_Q|_\infty \leq \frac{1}{2}\ell(Q)$  and  $|x - y|_\infty \leq \ell(Q)$ . Thus,  $|y - c_Q|_\infty \leq \frac{3}{2}\ell(Q)$ . In other words,  $y \in 3Q$ . Thus,

$$\chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right) = \chi_Q(x) \chi_{3Q}(y) \psi\left(\frac{x - y}{\ell(Q)}\right).$$

This completes the proof.  $\square$

**Lemma 4.3.** *Let  $r > 2$ . In the setting of Theorem 4.1, we have*

$$f(x)K(x, y) = \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k, l, Q}(x) f_{k, l, Q}(y),$$

where

(i)  $e_{k,l,Q}$  is supported in  $Q$  and  $f_{k,l,Q}$  is supported in  $3Q$  for every  $Q \in \mathbb{D}$  with  $Q \subset \Omega$ ;

(ii) for every  $k, l \in \mathbb{Z}^N$  and for every  $Q \in \mathbb{D}$  with  $Q \subset \Omega$ , we have

$$\|e_{k,l,Q}\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} = \|f\|_{L_r(Q, \mathbf{w}(x)dx)};$$

(iii) for every  $k, l \in \mathbb{Z}^N$  and for every  $Q \in \mathbb{D}$  with  $Q \subset \Omega$ , we have

$$\|f_{k,l,Q}\|_{L_\infty(\mathbb{R}^N)} \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)};$$

*Proof.* For every  $x, y \in \Omega$ , we have

$$\begin{aligned} K(x, y) &= \sum_{Q \in \mathbb{D}} \chi_Q(x) \chi_{3Q}(y) K(x, y) \psi\left(\frac{x-y}{\ell(Q)}\right) = \\ &= \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \chi_Q(x) \chi_{3Q}(y) K(x, y) \psi\left(\frac{x-y}{\ell(Q)}\right) = \end{aligned}$$

For every  $Q \in \mathbb{D}$  with  $3Q \subset \Omega$ , define the map

$$K_Q : \left[-\frac{1}{2}, \frac{1}{2}\right]^N \times \left[-\frac{3}{2}, \frac{3}{2}\right]^N \rightarrow \mathbb{C}$$

by setting

$$K_Q(a, b) = K(c_Q + a\ell(Q), c_Q + b\ell(Q))\psi(a-b), \quad a \in \left[-\frac{1}{2}, \frac{1}{2}\right]^N, \quad b \in \left[-\frac{3}{2}, \frac{3}{2}\right]^N.$$

If  $Q \in \mathbb{D}$  is such that  $3Q \not\subset \Omega$ , then the definition of  $K_Q$  is the same,  $a$  also runs through  $[-\frac{1}{2}, \frac{1}{2}]^N$ , but the parallelepiped  $b$  runs through is slightly smaller. **to co-authors: if you wish, specify the domain of  $b$  also.** In what follows, we lighten the notations by pretending that  $b$  is always defined on  $[-\frac{3}{2}, \frac{3}{2}]^N$ .

It follows from the assumption on  $K$  that

$$\|K_Q\|_{C^{2N+2}([-\frac{3}{2}, \frac{3}{2}]^N \times [-\frac{3}{2}, \frac{3}{2}]^N)} \leq \frac{c'_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

Hence, there exists an extension  $K_Q : \mathbb{T}^N \times \mathbb{T}^N \rightarrow \mathbb{C}$  such that

$$\|K_Q\|_{C^{2N+2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c''_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

In particular,

$$\|K_Q\|_{W^{2N+2,2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

We may, therefore, write

$$K_Q(a, b) = \sum_{k,l \in \mathbb{Z}^N} \text{coef}_{k,l,Q} e^{i\langle k,a \rangle} e^{i\langle l,b \rangle}, \quad a, b \in \mathbb{T}^N,$$

where

$$\sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{2N+2} |\text{coef}_{k,l,Q}|^2 \leq \left( \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)} \right)^2.$$

In particular,

$$(1 + |k|^2 + |l|^2)^{N+1} \cdot |\text{coef}_{k,l,Q}| \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}, \quad k, l \in \mathbb{Z}^N.$$

We now set

$$e_{k,l,Q}(x) = f(x)\chi_Q(x)e^{i\langle k, \frac{x-c_Q}{\ell(Q)} \rangle},$$

$$f_{k,l,Q}(y) = (1 + |k|^2 + |l|^2)^{N+1} \text{coef}_{k,l,Q} \cdot \chi_{3Q}(y)e^{i\langle l, \frac{y-c_Q}{\ell(Q)} \rangle}.$$

This allows us to write

$$f(x)K(x,y) = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k,l,Q}(x) f_{k,l,Q}(y).$$

Obviously,

$$\begin{aligned} \text{supp}(e_{k,l,Q}) &\subset Q, \quad \text{supp}(f_{k,l,Q}) \subset 3Q, \\ \|e_{k,l,Q}\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} &= \|f\|_{L_r(Q, \mathbf{w}(x)dx)}, \\ \|f_{k,l,Q}\|_{L_\infty(\mathbb{R}^N)} &= (1 + |k|^2 + |l|^2)^{N+1} |\text{coef}_{k,l,Q}| \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}. \end{aligned}$$

□

**Lemma 4.4.** *Let  $r > 2$ . In the setting of Theorem 4.1, we have*

$$\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq c_{K,\mathbf{w},r} \left\| \left\{ m(Q)^{\frac{1}{N}-\frac{1}{r}} \|f\|_{L_r(Q)} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \right\|_{N,\infty}.$$

*Proof.* Let  $A_{k,l}$  be the integral operator with integral kernel given by the formula

$$(x,y) \rightarrow \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k,l,Q}(x) f_{k,l,Q}(y), \quad x, y \in \Omega, \quad k, l \in \mathbb{Z}^N.$$

By Lemma 4.3, we have

$$M_f V = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} A_{k,l}.$$

By the triangle inequality,

$$\begin{aligned} &\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, w_\theta(x)dx))} \leq \\ &\leq \frac{N}{N-1} \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \|A_{k,l}\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, w_\theta(x)dx))}. \end{aligned}$$

By Lemma 4.3, the sequences

$$\left\{ \mathbf{w}(Q)^{\frac{1}{r}-\frac{1}{2}} \frac{e_{k,l,Q}}{\|f\|_{L_r(Q, \mathbf{w}(x)dx)}} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}}, \quad \left\{ \frac{c_{K,\theta}^{-1} \mathbf{w}(Q)^{\frac{1}{2}}}{\ell(Q)} f_{k,l,Q} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}}$$

are  $(r, 1)$ -NWO and, respectively,  $(\infty, 3)$ -NWO in the sense of Definition 2.4. Using Lemma 2.5, we conclude that

$$\begin{aligned} &\|A_{k,l}\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq \\ &\leq c_{\theta,r,1,\infty,3,N} c_{K,\theta} \left\| \left\{ \ell(Q) \mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \right\|_{N,\infty}. \end{aligned}$$

Therefore,

$$\begin{aligned} &\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}_+^N, \mathbf{w}(x)dx))} \leq \\ &\leq \frac{N}{N-1} c_{\theta,r,1,\infty,3,N} c_{K,\theta} \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \times \end{aligned}$$

$$\times \left\| \left\{ \ell(Q) \mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{N, \infty}.$$

Obviously,

$$\mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \leq \left( \frac{1}{\mathbf{w}(Q)} \cdot \sup_{x \in Q} \mathbf{w}(x) \right)^{\frac{1}{r}} \times \|f\|_{L_r(Q)}, \quad Q \in \mathbb{D}, \quad Q \subset \Omega.$$

Combining the last two estimates, we complete the proof.  $\square$

*Proof of Theorem 4.1.* Take  $r \in (2, N)$ . The assertion follows by combining Lemmas 4.4 and A.1.  $\square$

## 5. PROOF OF SPECIAL CBIKEL ESTIMATE IN DUNKL SETTING

**Lemma 5.1.** *For every pair of multi-indices  $\alpha, \beta \in \mathbb{Z}_+^N$ , for every  $t > 0$ , we have*

$$|(\partial_x^\alpha \partial_y^\beta h_t)(x, y)| \leq c_{\kappa, \alpha, \beta} t^{-\frac{|\alpha|_1 + |\beta|_1}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2).$$

*Proof.* This is (slightly weakened form of) Theorem 3.1 in [2].  $\square$

**Lemma 5.2.** *Suppose  $s + N > 2$ . We have*

$$\int_0^\infty t^{-\frac{s}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt \leq c_{\kappa, s} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s}.$$

*Proof.* By (2),

$$w(B(x, t^{\frac{1}{2}}))^{-1} \leq c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times \begin{cases} \left( \frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N, & 0 < t \leq d(x, y)^2, \\ \left( \frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N, & t \geq d(x, y)^2. \end{cases}$$

Therefore,

$$\begin{aligned} & \int_0^\infty t^{-\frac{s}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt \leq \\ & \leq c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times \int_0^{d(x, y)^2} t^{-\frac{s}{2}} \left( \frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt + \\ & \quad + c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times \int_{d(x, y)^2}^\infty t^{-\frac{s}{2}} \left( \frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N dt = \\ & = c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s} \times \int_0^1 t^{-\frac{s+N}{2}} \exp(-\frac{c_\kappa}{t}) dt + \\ & \quad + c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s} \times \int_1^\infty t^{-\frac{s+N}{2}} dt. \end{aligned}$$

$\square$

The following lemma establishes derivative estimates for the integral kernels of Dunkl-Riesz transforms.

**Lemma 5.3.** *Let  $N \geq 2$ . For every pair of multi-indices  $\alpha, \beta \in \mathbb{Z}_+^n$ , we have*

$$|(\partial_x^\alpha \partial_y^\beta K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}})(x, y)| \leq c_{\kappa, \alpha, \beta} \frac{d(x, y)^{1-|\alpha|_1-|\beta|_1}}{w(B(x, d(x, y)))}, \quad x, y \in \mathbb{R}^N.$$

*Proof.* We have

$$\Delta_{\text{Dunkl}}^{-\frac{1}{2}} = \pi^{-\frac{1}{2}} \int_0^\infty e^{-t\Delta_{\text{Dunkl}}} \frac{dt}{t^{\frac{1}{2}}}.$$

Hence,

$$K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}}(x, y) = \pi^{-\frac{1}{2}} \int_0^\infty h_t(x, y) \frac{dt}{t^{\frac{1}{2}}}.$$

Using Lemma 5.1, we estimate

$$\begin{aligned} & |(\partial_x^\alpha \partial_y^\beta K_{R_{\text{Dunkl},j}})(x, y)| \leq \\ & \leq \frac{c_{\kappa, \alpha, \beta}}{\pi^{\frac{1}{2}}} \int_0^\infty t^{-\frac{|\alpha|_1 + |\beta|_1 + 1}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt. \end{aligned}$$

Applying Lemma 5.2 with  $s = 1 + |\alpha|_1 + |\beta|_1$ , we conclude

$$\begin{aligned} & |(\partial_x^\alpha \partial_y^\beta K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}})(x, y)| \leq \\ & \leq \pi^{-\frac{1}{2}} c_{\kappa, \alpha, \beta} \times c_{\kappa, |\alpha|_1 + |\beta|_1 + 1} w(B(x, d(x, y)))^{-1} \times d(x, y)^{1 - |\alpha|_1 - |\beta|_1}. \end{aligned}$$

The assertion follows immediately.  $\square$

**Lemma 5.4.** *For every  $Q \in \mathbb{D}$  with  $Q \subset \Omega$ , we have*

$$\sup_{x \in Q} w_\theta(x) \leq c_\theta \frac{w_\theta(Q)}{m(Q)}.$$

*Proof.* We have

$$w(Q) \geq c_\theta \prod_{A \subset \{1, \dots, N\}} (\ell_Q + \sum_{k \in A} (c_Q)_k)^{\theta_A}.$$

On the other hand, if  $x \in Q$ , then  $|x - c_Q|_\infty \leq \frac{1}{2} \ell(Q)$  and, therefore,

$$\sum_{k \in A} x_k \leq \frac{|A|}{2} \ell_Q + \sum_{k \in A} (c_Q)_k \leq N(\ell_Q + \sum_{k \in A} (c_Q)_k).$$

Thus,

$$w(x) = \prod_{A \subset \{1, \dots, N\}} (\sum_{k \in A} x_k)^{\theta_A} \leq N^{\sum_{A \subset \{1, \dots, N\}} \theta_A} \prod_{A \subset \{1, \dots, N\}} (\ell_Q + \sum_{k \in A} (c_Q)_k)^{\theta_A}.$$

This completes the proof.  $\square$

**Theorem 5.5.** *Let  $N \geq 3$ . We have*

$$\|M_f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{s}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_\kappa \|f\|_{L_N(\mathbb{R}^N)}.$$

*Proof.* It suffices to estimate, for every chamber  $\Gamma$  and for every  $g \in G$ , the operator  $M_{\chi_\Gamma \cdot f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}} M_{\chi_{g\Gamma}}$ . In other words, it suffices to estimate an operator with integral kernel

$$f(x) K_{\text{Dunkl}^{-\frac{1}{2}}}(x, y), \quad x, y \in \Gamma, \quad y \in g\Gamma,$$

on the Hilbert space  $L_2(\mathbb{R}^N, w(x)dx)$ . In other words, it suffices to estimate an operator with integral kernel

$$f(x) K_{\text{Dunkl}^{-\frac{1}{2}}}(x, gy), \quad x, y \in \Gamma,$$

on the Hilbert space  $L_2(\Gamma, w(x)dx)$ .

Let  $N_1 = \dim(\text{span}(R))$ ,  $N_2 = N - N_1$  and let  $\Omega = \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ . For a concrete chamber  $\Gamma$ , let  $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$  be a linear isomorphism given in Theorem 2.2 so that  $\Phi(\Gamma) = \Omega$ .

It suffices to estimate an operator with integral kernel

$$f(\Phi^{-1}x)K_{\text{Dunkl}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space  $L_2(\Omega, w(\Phi^{-1}x)dx)$ . Using Lemma 3.5, it suffices to estimate an operator with integral kernel

$$f(\Phi^{-1}x)K_{\text{Dunkl}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space  $L_2(\Omega, w_\theta(x)dx)$ .

By Lemma 5.3,

$$\left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y) \right| \leq c_{\alpha, \beta, \kappa} \frac{|x - y|^{1 - |\alpha|_1 - |\beta|_1}}{w_\theta(B(x, |x - y|))}, \quad x, y \in \Omega.$$

The other assumptions in Theorem 4.1 follow from Lemma 5.4, Lemma 3.3 and Lemma 3.4. The assertion follows now from Theorem 4.1.  $\square$

## 6. PROOF OF THEOREM 1.1 (i)

**Lemma 6.1.** *We have*

$$[T_{e_j}, M_f] = M_{\partial_j f} + \frac{1}{2} \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} U_v,$$

where

$$(\delta_v f)(x) = \frac{f(x) - (U_v f)(x)}{\langle v, x \rangle}, \quad v \in R, \quad x \in \mathbb{R}^N.$$

*Proof.* Obvious.  $\square$

**Lemma 6.2.** *We have*

$$\|\delta_v f\|_{L_N(\mathbb{R}^N)} \leq c_N \|f\|_{\dot{W}^{1, N}(\mathbb{R}^N)}.$$

*Proof.* Without loss of generality,  $v = e_N$ . Set

$$(D_t h)(x) = h(x_1, \dots, x_{N-1}, tx_N).$$

We have

$$\delta_v f = \int_{-1}^1 D_t(\partial_N f) dt.$$

Thus,

$$\begin{aligned} \|\delta_v f\|_{L_N(\mathbb{R}^N)} &\leq \int_{-1}^1 \|D_t(\partial_N f)\|_{L_N(\mathbb{R}^N)} dt = \\ &= \int_{-1}^1 |t|^{-\frac{1}{N}} \|\partial_N f\|_{L_N(\mathbb{R}^N)} dt = \frac{2N}{N-1} \|\partial_N f\|_{L_N(\mathbb{R}^N)}. \end{aligned}$$

$\square$

**Lemma 6.3.** *For every  $N > 1$ , we have*

$$\|[B, A]B^{-1}\|_{N, \infty} \leq c_N \|B^{-1}[B^2, A]B^{-1}\|_{N, \infty}.$$

*Proof of Theorem 1.1 (i).* We prove the assertion under *a priori* restriction  $f \in C_c^\infty(\mathbb{R}^N)$ . Passing to the case of  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$  is standard (see e.g. [refer to our papers](#)).

We have

$$\begin{aligned} [R_{\text{Dunkl},j}, M_f] &= [T_{e_j} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}, M_f] = [T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} + T_{e_j} [\Delta_{\text{Dunkl}}^{-\frac{1}{2}}, M_f] = \\ &= [T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} - R_{\text{Dunkl},j} [\Delta_{\text{Dunkl}}^{\frac{1}{2}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}. \end{aligned}$$

By triangle inequality,

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|[T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} \|[\Delta_{\text{Dunkl}}^{\frac{1}{2}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Using Lemma 6.3, we write

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|[T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} c_N \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Note that

$$\begin{aligned} \Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} &= \\ &= \sum_{k=1}^N \Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f] R_{\text{Dunkl},k} + R_{\text{Dunkl},k} [T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}. \end{aligned}$$

By triangle inequality,

$$\begin{aligned} \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \sum_{k=1}^N \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} \sum_{k=1}^N \|[T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Combining the estimates in the last two paragraphs, we obtain

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq c'_N \sum_{k=1}^N \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ c'_N \sum_{k=1}^N \|[T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Using Lemma 6.1, the fact that  $\Delta_{\text{Dunkl}}$  commutes with  $O_v$  and the triangle inequality, we obtain

$$\begin{aligned} \|[T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|M_{\partial_k} f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \end{aligned}$$

$$+ \frac{N}{2(N-1)} \sum_{v \in R} |v| \kappa(v) \|M_{\delta_v f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}.$$

Thus,

$$\begin{aligned} & \| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ & \leq c_N'' \sum_{k=1}^N \|M_{\partial_k f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ & + c_\kappa \sum_{v \in R} \|M_{\delta_v f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

The assertion follows now from Theorem 5.5 and Lemma 6.2.  $\square$

## 7. PROOF OF THEOREM 1.1 (ii)

**Lemma 7.1.** *Let  $1 \leq j \leq N$ . If  $x_j \geq y_j$ , then*

$$K_{\text{Dunkl},j}(x, y) \geq c_\kappa \frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

*If  $x_j \leq y_j$ , then*

$$K_{\text{Dunkl},j}(x, y) \leq c_\kappa \frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

*Proof.* The proof is identical to that in [6, Theorem 1.2].  $\square$

**Lemma 7.2.** *Let  $Q \in \mathbb{D}$ . We have*

$$K_j(x, y) \geq \frac{c_\kappa}{w(Q)}, \quad x \in 3Q, \quad y \in Q + 3s(Q)e_j.$$

*Proof.* The assertion follows immediately from Lemma 7.1.  $\square$

**Lemma 7.3.** *Let real-valued  $f \in L_1^{\text{loc}}(\mathbb{R}^N)$  be such that the commutator  $[R_{\text{Dunkl},j}, M_f]$  is bounded on  $L_2(\mathbb{R}^N, w(x)dx)$ . For every  $Q \in \mathbb{D}$ , there exist  $\xi_Q$  and  $\eta_Q$  supported in  $7Q$  such that  $\|\xi_Q\|_\infty, \|\eta_Q\|_\infty \leq m(Q)^{-\frac{1}{2}}$  and such that*

$$\frac{1}{w(3Q)} \int_{3Q} |f(y) - f_{3Q}| w(y) dy \leq c_\kappa |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

*Proof.* Let  $\alpha$  be the median of  $f$  on  $3Q$  and  $\beta$  be the median of  $f$  on  $Q + 3s(Q)e_j$ . Assume for definiteness  $\alpha \geq \beta$ . Set

$$\begin{aligned} A_1 &= \{x \in 3Q : f(x) \geq \alpha\}, \quad B_1 = \{x \in Q + 3s(Q)e_j : f(x) \leq \beta\}, \\ A_2 &= \{x \in 3Q : f(x) \leq \beta\}, \quad B_2 = \{x \in Q + 3s(Q)e_j : f(x) \geq \alpha\}, \\ \xi_{Q,s} &= w(Q)^{-\frac{1}{2}} \chi_{A_s}, \quad \eta_{Q,s} = w(Q)^{-\frac{1}{2}} \chi_{B_s}, \quad s = 1, 2. \end{aligned}$$

We have

$$\begin{aligned} & \langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,s}, \eta_{Q,s} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} = \\ & = \frac{1}{w(Q)} \int_{B_s \times A_s} K_j(x, y) (f(y) - f(x)) w(x) w(y) dx dy. \end{aligned}$$

If  $x \in B_1$  and  $y \in A_1$ , then

$$\begin{aligned} f(y) - f(x) &= (f(y) - \alpha) + (\alpha - \beta) + (\beta - f(x)) = \\ &= |f(y) - \alpha| + |\alpha - \beta| + |f(x) - \beta| \geq |f(y) - \alpha| + |\alpha - \beta| \geq 0. \end{aligned}$$

If  $x \in B_2$  and  $y \in A_2$ , then

$$\begin{aligned} f(y) - f(x) &= (f(y) - \beta) + (\beta - f(x)) = \\ &= -|f(y) - \beta| - |f(x) - \beta| \leq -|f(y) - \beta| \leq 0. \end{aligned}$$

Thus,

$$\begin{aligned} &\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq \\ &\geq \frac{1}{w(Q)} \int_{B_1 \times A_1} K_j(x, y) (|f(y) - \alpha| + |\alpha - \beta|) w(x) w(y) dx dy \geq \\ &\geq \frac{c_\kappa}{w^2(Q)} \int_{B_1 \times A_1} (|f(y) - \alpha| + |\alpha - \beta|) w(x) w(y) dx dy = \\ &= \frac{c_\kappa}{w^2(Q)} \cdot \left( w(B_1) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(B_1) w(A_1) \right) \geq \\ &= \frac{c_\kappa}{w^2(Q)} \cdot \left( \frac{1}{2} w(Q) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| \cdot \frac{1}{4} w^2(Q) \right) \geq \\ &\geq \frac{c_\kappa}{4w(Q)} \cdot \left( \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(Q) \right) \end{aligned}$$

and

$$\begin{aligned} &-\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq \\ &\geq \frac{1}{w(Q)} \int_{B_2 \times A_2} K_j(x, y) |f(y) - \beta| w(x) w(y) dx dy \geq \\ &\geq \frac{c_\kappa}{w^2(Q)} \int_{B_2 \times A_2} |f(y) - \beta| w(x) w(y) dx dy = \\ &= \frac{c_\kappa}{w^2(Q)} \cdot w(B_2) \cdot \int_{A_2} |f(y) - \beta| w(y) dy \geq \\ &= \frac{c_\kappa}{2w(Q)} \int_{A_2} |f(y) - \beta| w(y) dy. \end{aligned}$$

It remains to note that

$$\begin{aligned} &\int_{3Q} |f(y) - \alpha| w(y) dy = \int_{A_1} |f(y) - \alpha| w(y) dy + \\ &+ \int_{A_2} |f(y) - \alpha| w(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |f(y) - \alpha| w(y) dy \leq \\ &\leq \int_{A_1} |f(y) - \alpha| w(y) dy + \int_{A_2} |f(y) - \beta| w(y) dy \\ &+ \int_{A_2} |\alpha - \beta| w(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |\alpha - \beta| w(y) dy \leq \\ &\leq \int_{A_1} |f(y) - \alpha| w(y) dy + \int_{A_2} |f(y) - \beta| w(y) dy + 2|\alpha - \beta| w(3Q). \end{aligned}$$

Combining the last 3 equations, we conclude

$$\frac{c'_\kappa}{w(Q)} \int_{3Q} |f(y) - \alpha| w(y) dy \leq \max_{s=1,2} |\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,s}, \eta_{Q,s} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

If

$$|\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}| \geq |\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|,$$

then we set  $\xi_Q = \xi_{Q,1}$  and  $\eta = \eta_{Q,1}$ . Otherwise, set  $\xi_Q = \xi_{Q,2}$  and  $\eta_Q = \eta_{Q,2}$ . Thus,

$$\frac{c'_\kappa}{w(Q)} \int_{3Q} |f(y) - \alpha| w(y) dy \leq |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

Hence,

$$c'_\kappa |f_{3Q} - \alpha| \leq |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|$$

and, finally,

$$\frac{c'_\kappa}{w(Q)} \int_{3Q} |f(y) - f_{3Q}| w(y) dy \leq 2 |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

□

**Lemma 7.4.** *Suppose that  $f \in L_1^{\text{loc}}(\mathbb{R}^N)$ . If  $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$  for some  $j \in \{1, 2, \dots, N\}$ , then*

$$\begin{aligned} & \left\| \left\{ \frac{1}{w(3Q)} \int_{3Q} |f(x) - f_{3Q}| w(x) dx \right\}_{Q \in \mathbb{D}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq \\ & \leq c_\kappa \| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

*Proof.* Let  $\{\xi_Q\}_{Q \in \mathbb{D}}$  and  $\{\eta_Q\}_{Q \in \mathbb{D}}$  be functions constructed in Lemma 7.3. By Lemma 2.6,

$$\left\{ \langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \right\}_{Q \in \mathbb{D}} \in l_p.$$

The first assertion follows now from Lemma 7.3. □

**Lemma 7.5.** *Let  $Q$  be a cube in  $\mathbb{R}^N$  (with sides parallel to coordinate hyperplanes). We have*

$$\sup_{x \in Q} w(x) \leq \frac{c_\kappa}{m(Q)} \int_Q w(x) dx.$$

*Proof.* If  $x = c_Q + y$ ,  $y \in [-\frac{1}{2}s(Q), \frac{1}{2}s(Q)]^d$ , then

$$|\langle x, v \rangle| \leq |\langle c_Q, v \rangle| + r |\langle y, v \rangle| \leq |\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q).$$

Thus,

$$w(x) = \prod_{v \in R} |\langle x, v \rangle|^{k(v)} \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q))^{k(v)}.$$

In particular,

$$LHS \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q))^{k(v)}.$$

Fix chamber  $\Gamma$  such that  $c_Q \in \Gamma$ . We have

$$\int_Q w(x) dx \geq \int_{c_Q + (B(0, \frac{1}{2}s(Q)) \cap \Gamma)} w(x) dx = \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma} w(c_Q + y) dy.$$

If  $y \in \Gamma$ , then

$$|\langle c_Q + y, v \rangle| = |\langle c_Q, v \rangle| + |\langle y, v \rangle|, \quad v \in R.$$

Thus,

$$w(c_Q + y) = \prod_{v \in R} (|\langle c_Q, v \rangle| + |\langle y, v \rangle|)^{k(v)}.$$

Hence,

$$\int_Q w(x)dx \geq \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma} \prod_{v \in R} (|\langle c_Q, v \rangle| + |\langle y, v \rangle|)^{k(v)} dy.$$

Fix a closed cone  $\Gamma' \subset \Gamma$  such that (i) interior of  $\Gamma'$  is non-empty and (ii)  $\partial\Gamma \cap \partial\Gamma' = \{0\}$ . We have

$$|\langle y, v \rangle| \geq c_{\Gamma'} |y|, \quad y \in \Gamma', \quad v \in R.$$

We, therefore, have

$$\begin{aligned} \int_Q w(x)dx &\geq \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma'} \prod_{v \in R} (|\langle c_Q, v \rangle| + c_{\Gamma'} |y|)^{k(v)} dy \geq \\ &\geq \int_{\frac{1}{4}s(Q) \leq |y| \leq \frac{1}{4}s(Q)} \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{1}{2}c_{\Gamma'} s(Q))^{k(v)} dy = \\ &= s(Q)^N \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{1}{4}c_{\Gamma'} s(Q))^{k(v)} \times (2^{-N} - 4^{-N}) \int_{\substack{y \in \Gamma' \\ |y| \leq 1}} dy. \end{aligned}$$

Combining this with the preceding paragraph, we complete the proof.  $\square$

**Lemma 7.6.** Suppose  $1 < t < 1 + \frac{1}{|R| \cdot \max_{v \in R} \kappa(v)}$ . Let  $Q$  be a cube in  $\mathbb{R}^N$  (with sides parallel to coordinate hyperplanes). We have

$$\left( \frac{\int_Q w^{1-t}(x)dx}{\int_Q w(x)dx} \right)^{\frac{1}{t}} \leq c_{t,\kappa} \frac{m(Q)}{\int_Q w(x)dx}.$$

*Proof.* If  $v \in R$  is such that

$$\sqrt{d}s(Q) \leq |\langle c_Q, v \rangle|,$$

then

$$|\langle x, v \rangle| \geq |\langle c_Q, v \rangle| - |\langle x - c_Q, v \rangle| \geq |\langle c_Q, v \rangle| - \frac{\sqrt{d}}{2}s(Q) \geq \frac{1}{2}|\langle c_Q, v \rangle|, \quad x \in Q.$$

In this case,

$$\begin{aligned} \int_Q |\langle x, v \rangle|^{(1-t)|R|\kappa(v)} dx &\leq 2^{(t-1)|R|\kappa(v)} |\langle c_Q, v \rangle|^{(1-t)|R|\kappa(v)} \cdot \int_Q dx = \\ &= 2^{(t-1)|R|\kappa(v)} |\langle x_0, v \rangle|^{(1-t)|R|\kappa(v)} \cdot s(Q)^N. \end{aligned}$$

If  $v \in R$  is such that

$$|\langle c_Q, v \rangle| \leq \sqrt{d}s(Q),$$

then let  $y_0 = \langle c_Q, v \rangle$  and  $z_0 = c_Q - y_0 v$ . Let us introduce new variables  $y = \langle x, v \rangle$  and  $z = x - yv$  (it varies in the plane orthogonal to  $v$ ). We write

$$\begin{aligned} \int_Q |\langle x, v \rangle|^{(1-t)|R|\kappa(v)} dx &\leq \int_{|x - c_Q| \leq \frac{\sqrt{d}}{2}s(Q)} |\langle x, v \rangle|^{(1-t)|R|\kappa(v)} dx \leq \\ &\leq \int_{|y - y_0|, |z - z_0| \leq \frac{\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|\kappa(v)} dy dz = \\ &= \left( \frac{\sqrt{d}}{2}s(Q) \right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y - y_0| \leq \frac{\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|\kappa(v)} dy \leq \\ &= \left( \frac{\sqrt{d}}{2}s(Q) \right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y| \leq \frac{3\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|\kappa(v)} dy = \end{aligned}$$

$$= s(Q)^{N+(1-t)|R|\kappa(v)} \cdot \left(\frac{\sqrt{d}}{2}\right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y| \leq \frac{3\sqrt{d}}{2}} |y|^{(1-t)|R|\kappa(v)} dy.$$

Combining the estimates in the preceding paragraphs, we write

$$\int_Q |\langle x, v \rangle|^{(1-t)|R|\kappa(v)} dx \leq c_1 s(Q)^N (\max\{|\langle c_Q, v \rangle|, r\})^{(1-t)|R|\kappa(v)}.$$

By Hölder inequality,

$$\begin{aligned} \int_Q w^{1-t}(x) dx &\leq \prod_{v \in R} \left( \int_Q |\langle x, v \rangle|^{(1-t)|R|\kappa(v)} dx \right)^{\frac{1}{|R|}} \leq \\ &\leq c_2 s(Q)^N \prod_{v \in R} (\max\{|\langle c_Q, v \rangle|, s(Q)\})^{(1-t)\kappa(v)}. \end{aligned}$$

The assertion follows easily.  $\square$

**Lemma 7.7.** *Measures  $dx$  and  $w(x)dx$  (and distance  $(x, y) \rightarrow |x - y|_\infty$ ) satisfy the conditions in Proposition 1.2 in [9].*

*Proof.* That both measures are doubling is obvious.

Let  $Q$  be a cube in  $\mathbb{R}^N$  (with sides parallel to coordinate hyperplanes). By Lemmas 7.5 and 7.6,

$$\begin{aligned} \left( \frac{1}{m(Q)} \int_Q w^t(x) dx \right)^{\frac{1}{t}} &\leq \frac{c_\kappa}{m(Q)} \int_Q w(x) dx, \quad t > 0. \\ \left( \frac{\int_Q w^{1-t}(x) dx}{\int_Q w(x) dx} \right)^{\frac{1}{t}} &\leq c_G \frac{m(Q)}{\int_Q w(x) dx}, \quad 1 < t < 1 + \frac{1}{|R| \cdot \max_{v \in R} \kappa(v)}. \end{aligned}$$

The assertion follows now from Proposition 3.6 in [9].  $\square$

*Proof of Theorem 1.1 (ii).* The left hand side in Lemma 7.4 is the oscillation semi-norm of the function  $f$  with respect to measure  $w(x)dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$ . So, the oscillation semi-norm of the function  $f$  with respect to measure  $w(x)dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$  is controlled by the  $\|\cdot\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator  $[R_{\text{Dunkl},j}, M_f]$ .

Lemma 7.7 asserts that measures  $dx$  and  $w(x)dx$  (and distance  $(x, y) \rightarrow |x - y|_\infty$ ) satisfy the conditions in Proposition 1.2 in [9]. Appealing to Proposition 1.2 in [9], we conclude that the oscillation semi-norm of the function  $f$  with respect to measure  $dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$  is controlled by the one with respect to measure  $w(x)dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$ . By the preceding paragraph, the oscillation semi-norm of the function  $f$  with respect to measure  $dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$  is controlled by the  $\|\cdot\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator  $[R_{\text{Dunkl},j}, M_f]$ .

That oscillation semi-norm of the function  $f$  with respect to measure  $dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$  is equivalent to the Sobolev semi-norm of the function  $f$  is a standard fact. The proof (in much more general setting) is available in Proposition 1.29 in [10]. The assertion follows now from the preceding paragraph.  $\square$

## 8. PROOF OF SMOOTHNESS

**Theorem 8.1.** *Let  $f \in C_c^\infty(\mathbb{R}^N)$ .*

(i) *for every  $n \in \mathbb{N}$ ,  $s_1, s_2 \in \mathbb{R}$  with  $s_1 + s_2 = n$ , the operator<sup>1</sup>*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

*is bounded on  $L_2(\mathbb{R}^N, w(x)dx)$ ;*

(ii) *for every  $n \in \mathbb{N}$ ,  $s_1, s_2 \in \mathbb{R}$  with  $s_1 + s_2 > n$ , the operator<sup>2</sup>*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

*belongs to  $\mathcal{L}_{\frac{N}{s_1+s_2-n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ ;*

**Lemma 8.2.** *Let  $f \in C_c^\infty(\mathbb{R}^N)$ . We have*

$$[\Delta_{\text{Dunkl}}, M_f] = 2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} (T_{e_j} U_v + U_v T_{e_j}) + \sum_{g \in G} M_{f_g} U_g.$$

*All functions on the right hand side are smooth and compactly supported.*

*Proof.* We have

$$\begin{aligned} [\Delta_{\text{Dunkl}}, M_f] &= \sum_{j=1}^N T_{e_j} [T_{e_j}, M_f] + [T_{e_j}, M_f] T_{e_j} = \\ &= \sum_{j=1}^N T_{e_j} M_{\partial_j f} + M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (T_{e_j} M_{\delta_v f} U_v + M_{\delta_v f} U_v T_{e_j}) = \\ &= 2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (M_{\delta_v f} T_{e_j} U_v + M_{\delta_v f} U_v T_{e_j}) + \\ &\quad + \sum_{j=1}^N [T_{e_j}, M_{\partial_j f}] + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) [T_{e_j}, M_{\delta_v f}] U_v = \\ &= 2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (M_{\delta_v f} T_{e_j} U_v + M_{\delta_v f} U_v T_{e_j}) + \\ &\quad + \sum_{j=1}^N M_{\partial_j^2 f} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} \alpha_j \kappa(v) M_{\delta_v(\partial_j f)} U_v + \\ &\quad + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) M_{\partial_j(\delta_v f)} U_v + \frac{1}{4} \sum_{j=1}^N \sum_{v, v' \in R} v_j v'_j \kappa(v) \kappa(v') M_{\delta_{v'} \delta_v f} U_{v'} U_v. \end{aligned}$$

□

**Lemma 8.3.** *For every  $f \in C_c^\infty(\mathbb{R}^N)$ , the operator*

$$[\Delta_{\text{Dunkl}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$$

*is bounded and*

$$(1 + \Delta_{\text{Dunkl}})^{-1} [\Delta_{\text{Dunkl}}, M_f] \in \mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

<sup>1</sup>Commutators are repeated  $n$  times

<sup>2</sup>Commutators are repeated  $n$  times

*Proof.* We prove only the first assertion. The proof of the second one is similar (modulo Theorem 5.5).

Since each  $T_{e_j}$  commutes with  $(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$ , each  $U_v$  commutes with  $(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$  and since the operators

$$\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, \quad 1 \leq j \leq N,$$

are bounded, the assertion follows from Lemma 8.2.  $\square$

**Lemma 8.4.** *For every  $n \in \mathbb{N}$ , the operator<sup>3</sup>*

$$[\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]](1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}}$$

*is bounded.*

*Proof.* Denote for brevity

$$\text{Expr}_n(f) = [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]](1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}}.$$

We prove the assertion by induction on  $f$ . The  $n$ -th statement is "for every  $f \in C_c^\infty(\mathbb{R}^N)$ , the operator  $\text{Expr}_n(f)$  is bounded." Base of induction is given by Lemma 8.3. It suffices to prove the step of induction.

Since each  $T_{e_j}$  and each  $U_v$  commutes with  $\Delta_{\text{Dunkl}}$ , it follows from Lemma 8.2 that

$$\begin{aligned} \text{Expr}_{n+1}(f) &= 2 \sum_{j=1}^N \text{Expr}_n(\partial_j f) \cdot \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + \\ &+ \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) \text{Expr}_n(\delta_v f) \cdot \left( \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} U_v + U_v \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \right) + \\ &+ \sum_{g \in G} \text{Expr}_n(f_g) \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} U_g. \end{aligned}$$

In each summand, the first factor is bounded by inductive assumption and the second factor is obviously bounded. This yields the step of induction and, hence, completes the proof.  $\square$

*Proof of Theorem 8.1 (i).* The assertion follows from Lemma 8.4 and Hadamard 3 lines theorem.  $\square$

**Lemma 8.5.** *For every  $f \in C_c^\infty(\mathbb{R}^N)$ , we have*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-1} \in \mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

*Proof.* Set

$$\mathcal{D} = \sum_{j=1}^N \gamma_j \otimes T_{e_j},$$

where  $\{\gamma_j\}_{j=1}^N$  are Pauli matrices. We have

$$\begin{aligned} 1 \otimes M_f(1 + \Delta_{\text{Dunkl}})^{-1} &= (1 \otimes M_f)(\mathcal{D}^2 + 1)^{-1} = (1 \otimes M_f)(\mathcal{D} + i)^{-1}(\mathcal{D} - i)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + [1 \otimes M_f, (\mathcal{D} + i)^{-1}](\mathcal{D} - i)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + (\mathcal{D} + i)^{-1}[\mathcal{D}, 1 \otimes M_f](\mathcal{D}^2 + 1)^{-1} = \end{aligned}$$

---

<sup>3</sup>Commutators are repeated  $n$  times

$$\begin{aligned}
&= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + (\mathcal{D} + i)^{-1} \left( \sum_{j=1}^N \gamma_j \otimes [T_{e_j}, M_f] \right) (\mathcal{D}^2 + 1)^{-1} = \\
&= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + \sum_{j=1}^N (\mathcal{D} + i)^{-1} (\gamma_j \otimes M_{\partial_j f}) (\mathcal{D}^2 + 1)^{-1} + \\
&\quad + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (\mathcal{D} + i)^{-1} (\gamma_j \otimes M_{\delta_v f} U_v) (\mathcal{D}^2 + 1)^{-1}.
\end{aligned}$$

Since  $\Delta_{\text{Dunkl}}$  commutes with  $U_v$ , it follows that

$$\begin{aligned}
&1 \otimes M_f (1 + \Delta_{\text{Dunkl}})^{-1} = \\
&= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + \sum_{j=1}^N (\mathcal{D} + i)^{-1}(1 \otimes M_{\partial_j f})(\mathcal{D}^2 + 1)^{-1} \cdot (\gamma_j \otimes 1) + \\
&\quad + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (\mathcal{D} + i)^{-1}(1 \otimes M_{\delta_v f})(\mathcal{D}^2 + 1)^{-1} \cdot (\gamma_j \otimes U_v).
\end{aligned}$$

Clearly,

$$\begin{aligned}
&\|(\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} = \\
&= \|(\Delta_{\text{Dunkl}} + 1)^{-\frac{1}{2}} M_f (\Delta_{\text{Dunkl}} + 1)^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\
&= \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} M_f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\
&\leq 2^{\frac{2}{N}} \|M_{|f|^{\frac{1}{2}}} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))}^2 \leq \\
&\stackrel{\text{Th. 5.5}}{\leq} c_{N, \kappa} \| |f|^{\frac{1}{2}} \|_{L_N(\mathbb{R}^N)}^2 = c_{N, \kappa} \|f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)}
\end{aligned}$$

and, similarly,

$$\begin{aligned}
&\|(\mathcal{D} + i)^{-1}(1 \otimes M_{\partial_j f})(\mathcal{D}^2 + 1)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} \leq c_{N, \kappa} \|\partial_j f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)}, \\
&\|(\mathcal{D} + i)^{-1}(1 \otimes M_{\delta_v f})(\mathcal{D}^2 + 1)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} \leq c_{N, \kappa} \|\delta_v f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)}.
\end{aligned}$$

This completes the proof.  $\square$

**Lemma 8.6.** *For every  $f \in C_c^\infty(\mathbb{R}^N)$ , we have*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \in \mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

*Proof.* We prove the assertion by induction on  $f$ . The  $n$ -th statement is "for every  $f \in C_c^\infty(\mathbb{R}^N)$ , we have

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \in \mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))."$$

Base of induction (i.e. cases  $n = 1$  and  $n = 2$ ) are given by Theorem 5.5 and Lemma 8.5. It suffices to prove the step of induction.

We have

$$\begin{aligned}
&M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}} = \\
&= (1 + \Delta_{\text{Dunkl}})^{-1} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + [M_f, (1 + \Delta_{\text{Dunkl}})^{-1}] (1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} = \\
&= (1 + \Delta_{\text{Dunkl}})^{-1} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + (1 + \Delta_{\text{Dunkl}})^{-1} [\Delta_{\text{Dunkl}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}}.
\end{aligned}$$

Choose  $\phi \in C_c^\infty(\mathbb{R}^N)$  such that  $(f \circ U_g) \cdot \phi = f \circ U_g$  for every  $g \in G$ . It follows that

$$[\Delta_{\text{Dunkl}}, M_f] = [\Delta_{\text{Dunkl}}, M_f] M_\phi.$$

Thus,

$$\begin{aligned} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}} &= (1 + \Delta_{\text{Dunkl}})^{-1} M_f \cdot M_\phi(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + \\ &+ (1 + \Delta_{\text{Dunkl}})^{-1} [\Delta_{\text{Dunkl}}, M_f] \cdot M_\phi(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}. \end{aligned}$$

In the first summand on the right hand side, the first factor belongs to the ideal  $\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$  by Lemma 8.5 and the second one belongs to the ideal  $\mathcal{L}_{\frac{N}{n-1}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$  by inductive assumption. In the second summand on the right hand side, the first factor belongs to the ideal  $\mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx))$  by Lemma 8.3, the second factor belongs to the ideal  $\mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$  by inductive assumption and the third one is obviously bounded.  $\square$

**Lemma 8.7.** *Let  $k, l \in \mathbb{Z}_+$  with  $l > k$ . For every  $f \in C_c^\infty(\mathbb{R}^N)$ , we have*

$$T_{e_{m_1}} \cdots T_{e_{m_k}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}$$

*belongs to  $\mathcal{L}_{\frac{N}{l-k}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ .*

*Proof.* We prove the assertion by induction on  $k$ . Base of induction (i.e., the case  $k = 0$ ) is established in Lemma 8.6.

We have

$$\begin{aligned} T_{e_{m_1}} \cdots T_{e_{m_k}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} &= T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_f T_{e_{m_k}}(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} + \\ &+ T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} [T_{e_{m_k}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} = \\ &= T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times \frac{T_{e_{m_k}}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + \\ &+ T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_{\partial_{m_k}} f(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + \\ &+ \frac{1}{2} \sum_{v \in R} v_{m_k} \kappa(v) T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_{\delta_v} f(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} U_v. \end{aligned}$$

In each summand on the right hand side, the first factor belongs to the ideal  $\mathcal{L}_{\frac{N}{l-k}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$  by inductive assumption and the second factor is obviously bounded. This yields the step of induction and, hence, completes the proof.  $\square$

**Lemma 8.8.** *Let  $s_1, s_2 \in \mathbb{R}$  be such that  $s_1 + s_2 > 0$ . We have*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

*belongs to  $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ .*

*Proof. Step 1:* Let us prove the assertion for  $s_1, s_2 \in \mathbb{Z}$ .

Suppose first  $s_1 \leq 0$ . Denote  $k = -s_1 \geq 0$  and  $l = s_2 > k$ . Denote for brevity

$$V = (1 + \Delta_{\text{Dunkl}})^{\frac{k}{2}} (1 + \sum_{j_1, \dots, j_k=1}^N |T_{e_{j_1}} \cdots T_{e_{j_k}}|)^{-1}.$$

Obviously,  $V$  is bounded. We have

$$\begin{aligned} (1 + \Delta_{\text{Dunkl}})^{\frac{k}{2}} &= V \times (1 + \sum_{m_1, \dots, m_k=1}^N |T_{e_{m_1}} \cdots T_{e_{m_k}}|) = V + \\ &+ \sum_{m_1, \dots, m_k=1}^N V \times \text{sgn}(T_{e_{m_1}} \cdots T_{e_{m_k}})^* \times T_{e_{m_1}} \cdots T_{e_{m_k}}. \end{aligned}$$

It, therefore, suffices to prove that operators

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}, \quad T_{e_{m_1}} \cdots T_{e_{m_k}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}$$

belong to  $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ . This is established in Lemma 8.6 and Lemma 8.7, respectively. Hence, the claim is proved for  $s_1, s_2 \in \mathbb{Z}$  and  $s_1 \leq 0$ .

If  $s_2 \leq 0$ , then the claim follows from that for  $s_2 \leq 0$  by taking the adjoint. If  $s_1, s_2 \geq 0$ , then, according to Lemma 8.7 the operators

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1+s_2}{2}} M_f, \quad M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1+s_2}{2}}$$

belong to  $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ . The claim for  $s_1, s_2 \geq 0$  follows now from the Hadamard 3 lines theorem.

**Step 2:** The general case follows now from Step 1 by complex interpolation.  $\square$

*Proof of Theorem 8.1 (ii).* Choose  $\phi \in C_c^\infty(\mathbb{R}^N)$  such that  $(f \circ U_g) \cdot \phi = f \circ U_g$  for every  $g \in G$ . It follows that

$$[\Delta_{\text{Dunkl}}, M_f] = [\Delta_{\text{Dunkl}}, M_f] M_\phi.$$

We have

$$\begin{aligned} &(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] M_\phi (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{n-s_1}{2}} \times \\ &\quad \times (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1-n}{2}} M_\phi (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}. \end{aligned}$$

The first factor is bounded by Theorem 8.1 (i). The second factor belongs to  $\mathcal{L}_{\frac{N}{s_1+s_2-n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$  by Lemma 8.8.  $\square$

## 9. APPROXIMATION THEOREM

**Theorem 9.1.** *For every  $f \in C_c^\infty(\mathbb{R}^N)$ , we have*

$$\begin{aligned} M_{w^{\frac{1}{2}}} [R_{\text{Dunkl}, j}, M_f] M_{w^{-\frac{1}{2}}} &\in V \Delta^{-\frac{1}{2}} + (\mathcal{L}_{N, \infty})_0(L_2(\mathbb{R}^N)), \\ V &= M_{\partial_j f} + \frac{1}{2} \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} U_v - \sum_{k=1}^N R_j M_{\partial_k f} R_k - \\ &\quad - \frac{1}{4} R_j \cdot \sum_{k=1}^N \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} (R_k U_v + U_v R_k). \end{aligned}$$

### 9.1. First approximation theorem.

**Theorem 9.2.** *For every  $f \in C_c^\infty(\mathbb{R}^N)$ , we have*

$$\begin{aligned} & [(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \frac{1}{2}[\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} \in \\ & \in \mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx)). \end{aligned}$$

**Lemma 9.3.** *Let  $B \geq 1$ . Suppose*

$$B^{-1}[B^2, [B^2, A]]B^{-\frac{5}{2}} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

*We have*

$$[B, A]B^{-1} - \frac{1}{2}[B^2, A]B^{-2} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

*Proof.* We have

$$B = \frac{2}{\pi} \int_0^\infty \frac{B^2 d\lambda}{\lambda^2 + B^2}.$$

Hence,

$$[B, A]B^{-1} = \frac{2}{\pi} \int_0^\infty \left[ \frac{B^2}{\lambda^2 + B^2}, A \right] B^{-1} d\lambda.$$

Obviously,

$$\begin{aligned} \left[ \frac{B^2}{\lambda^2 + B^2}, A \right] B^{-1} &= - \left[ \frac{\lambda^2}{\lambda^2 + B^2}, A \right] B^{-1} = \frac{\lambda^2}{\lambda^2 + B^2} [B^2, A] \frac{1}{B(\lambda^2 + B^2)} = \\ &= [B^2, A] \frac{\lambda^2}{B(\lambda^2 + B^2)^2} - \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2}. \end{aligned}$$

Obviously,

$$\begin{aligned} \int_0^\infty [B^2, A] \frac{\lambda^2}{B(\lambda^2 + B^2)^2} d\lambda &= [B^2, A] B^{-1} \cdot \int_0^\infty \frac{\lambda^2}{(\lambda^2 + B^2)^2} d\lambda = \\ &= [B^2, A] B^{-2} \cdot \int_0^\infty \frac{\lambda^2}{(\lambda^2 + 1)^2} d\lambda = \frac{\pi}{4} [B^2, A] B^{-2}. \end{aligned}$$

On the other hand,

$$\begin{aligned} & \left\| \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} \right\|_{\frac{2N}{3}, \infty} \leq \\ & \leq \left\| \frac{\lambda^2 B}{\lambda^2 + B^2} \right\|_\infty \|B^{-1}[B^2, [B^2, A]]B^{-\frac{5}{2}}\|_{\frac{2N}{3}, \infty} \left\| \frac{B^{\frac{3}{2}}}{(\lambda^2 + B^2)^2} \right\|_\infty. \end{aligned}$$

Next,

$$\left\| \frac{\lambda^2 B}{\lambda^2 + B^2} \right\|_\infty \leq \sup_{t \geq 1} \frac{\lambda^2 t}{\lambda^2 + t^2} \leq \lambda$$

and

$$\left\| \frac{B^{\frac{3}{2}}}{(\lambda^2 + B^2)^2} \right\|_\infty \leq \sup_{t \geq 1} \frac{t^{\frac{3}{2}}}{(\lambda^2 + t^2)^2} \leq (\lambda^2 + 1)^{-\frac{5}{4}}.$$

Thus,

$$\begin{aligned} & \left\| \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} \right\|_{\frac{2N}{3}, \infty} \leq \\ & \leq \frac{\lambda}{(\lambda^2 + 1)^{\frac{5}{4}}} \|B^{-1}[B^2, [B^2, A]]B^{-\frac{5}{2}}\|_{\frac{2N}{3}, \infty}. \end{aligned}$$

Hence, the integral

$$\int_0^\infty \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} d\lambda$$

converges absolutely in  $\mathcal{L}_{\frac{2N}{3}, \infty}$  and the assertion follows.  $\square$

*Proof of Theorem 9.2.* The assertion follows from Theorem 8.1 and Lemma 9.3.  $\square$

## 9.2. Second approximation theorem.

**Theorem 9.4.** *If  $f \in C_c^\infty(\mathbb{R}^N)$ , then*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \in (\mathcal{L}_{N, \infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

**Lemma 9.5.** *Let  $A, B_1, B_2$  be such that  $B_1, B_2 \geq 1$ . Suppose that*

- (i)  $A(B_2 - B_1)B_2^{-\frac{1}{2}}, [B_1, A]B_1^{-\frac{1}{2}}$  and  $[B_1, A]B_2^{-\frac{1}{2}}$  are bounded;
- (ii) *there exists  $A'$  with  $B_1^{-\frac{3}{4}}A' \in \mathcal{L}_{\frac{2N}{3}, \infty}$  such that  $A = A'A$  and  $[B_1, A] = A'[B_1, A]$ ;*

*We have*

$$AB_1^{-\frac{1}{2}} - AB_2^{-\frac{1}{2}} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

*Proof.* We have

$$B_k^{-\frac{1}{2}} = \frac{2}{\pi} \int_0^\infty \frac{d\lambda}{\lambda^2 + B_k}, \quad k = 1, 2.$$

Thus,

$$AB_1^{-\frac{1}{2}} - AB_2^{-\frac{1}{2}} = \int_0^\infty A \left( \frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_2} \right) d\lambda.$$

Obviously,

$$\begin{aligned} A \left( \frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_2} \right) &= A \frac{1}{\lambda^2 + B_1} (B_2 - B_1) \frac{1}{\lambda^2 + B_2} = \\ &= \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} + \frac{1}{\lambda^2 + B_1} [B_1, A] \frac{1}{\lambda^2 + B_1} (B_2 - B_1) \frac{1}{\lambda^2 + B_2} = \\ &= \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} + \frac{1}{\lambda^2 + B_1} [B_1, A] \frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_1} [B_1, A] \frac{1}{\lambda^2 + B_2}. \end{aligned}$$

Consider, for example, the first summand. We write

$$\begin{aligned} \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} &= \frac{1}{\lambda^2 + B_1} A' \cdot A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} = \\ &= \frac{B_1^{\frac{3}{4}}}{\lambda^2 + B_1} \cdot B_1^{-\frac{3}{4}} A' \cdot A(B_2 - B_1) B_2^{-\frac{1}{2}} \cdot \frac{B_2^{\frac{1}{2}}}{\lambda^2 + B_2}. \end{aligned}$$

Thus,

$$\begin{aligned} &\left\| \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} \right\|_{\frac{3N}{4}, \infty} \leq \\ &\leq \left\| \frac{B_1^{\frac{3}{4}}}{\lambda^2 + B_1} \right\|_\infty \cdot \left\| B_1^{-\frac{3}{4}} A' \right\|_{\frac{3N}{4}, \infty} \cdot \|A(B_2 - B_1) B_2^{-\frac{1}{2}}\|_\infty \cdot \left\| \frac{B_2^{\frac{1}{2}}}{\lambda^2 + B_2} \right\|_\infty \leq \\ &\leq \sup_{t \geq 1} \frac{t^{\frac{3}{4}}}{\lambda^2 + t} \cdot \left\| B_1^{-\frac{3}{4}} A' \right\|_{\frac{3N}{4}, \infty} \cdot \|A(B_2 - B_1) B_2^{-\frac{1}{2}}\|_\infty \cdot \sup_{t \geq 1} \frac{t^{\frac{1}{2}}}{\lambda^2 + t} \leq \\ &\leq (\lambda^2 + 1)^{-\frac{1}{4}} \cdot \left\| B_1^{-\frac{3}{4}} A' \right\|_{\frac{3N}{4}, \infty} \cdot \|A(B_2 - B_1) B_2^{-\frac{1}{2}}\|_\infty \cdot (\lambda^2 + 1)^{-\frac{1}{2}}. \end{aligned}$$

Hence, the first summand is absolutely integrable in  $\mathcal{L}_{\frac{3N}{4}, \infty}$ . Similarly, so are the other summands. This completes the proof.  $\square$

**Lemma 9.6.** *Let  $P$  be a differential operator of degree 1 with smooth compactly supported coefficients. Suppose that the coefficients are supported away from the hyperplanes  $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$ . The operators*

$$P(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}, \quad P(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}$$

*are bounded on  $L_2(\mathbb{R}^N, w(x)dx)$ .*

*Proof.* By assumption,  $P = M_{w^{-\frac{1}{2}}} P' M_{w^{\frac{1}{2}}}$ , where  $P'$  is a differential operator of degree 1 with smooth compactly supported coefficients. Thus,

$$P(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} = M_{w^{-\frac{1}{2}}} \cdot P'(1 + \Delta)^{-\frac{1}{2}} \cdot M_{w^{\frac{1}{2}}}.$$

Boundedness of the right hand side is obvious.

Similarly,

$$P = \sum_{j=1}^N M_{f_j} T_{e_j} + \sum_{g \in G} M_{h_g} U_g,$$

where functions  $\{f_j\}_{j=1}^N$  and  $\{h_g\}_{g \in G}$  are smooth and compactly supported. Since the operators  $\{U_g\}_{g \in G}$  commute with  $\Delta_{\text{Dunkl}}$ , it follows that

$$P(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} = \sum_{j=1}^N M_{f_j} \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + \sum_{g \in G} M_{h_g} \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} U_g.$$

Boundedness of the right hand side is obvious.  $\square$

*Proof of Theorem 9.4. Step 1:* Suppose first that  $f$  is supported away from the hyperplanes  $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$ .

We apply Lemma 9.5 with  $A = M_f$ ,  $B_1 = 1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}$  and  $B_2 = 1 + \Delta_{\text{Dunkl}}$ .

Choose  $\phi \in C_c^\infty(\mathbb{R}^N)$  such that  $f = f\phi$  and set  $A' = M_\phi$ . We have

$$M_\phi[M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}, M_f] = [M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}, M_f].$$

Thus,  $A = A'A$  and  $[B_1, A] = A'[B_1, A]$ . It follows from the usual Cwikel estimates that  $B_1^{-\frac{3}{4}} A' \in \mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ . This verifies the assumption (ii) in Lemma 9.5.

Note that  $[B_1, A]$  is a differential operator  $P$  of degree 1 with smooth compactly supported coefficients. Furthermore, the coefficients are supported away from the hyperplanes  $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$ . By Lemma 9.6, the operators

$$[B_1, A]B_1^{-\frac{1}{2}} = P(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}, \quad [B_1, A]B_2^{-\frac{1}{2}} = P(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}},$$

are bounded on  $L_2(\mathbb{R}^N, w(x)dx)$ . Note that there exist differential operators  $\{P_g\}_{g \in G}$  of degree 1 with smooth compactly supported coefficients such that the coefficients are supported away from the hyperplanes  $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$  and such that

$$A(B_2 - B_1) = \sum_{g \in G} P_g U_g.$$

Since  $\{U_g\}_{g \in G}$  commute with  $\Delta_{\text{Dunkl}}$ , it follows that

$$A(B_2 - B_1) = \sum_{g \in G} P_g (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} \cdot U_g$$

is bounded on  $L_2(\mathbb{R}^N, w(x)dx)$ . This verifies the assumption (i) in Lemma 9.5.

Using Lemma 9.5, we infer the claim of Step 1.

**Step 2:** Choose a sequence  $\{\phi_n\}_{n \geq 1} \subset C_c^\infty(\mathbb{R}^N)$  such that (a)  $0 \leq \phi_n \leq 1$  for  $n \geq 1$  (b)  $\phi_n$  is supported away from the hyperplanes  $\{x : \langle x, v \rangle = 0\}$ ,  $v \in R$ , for  $n \geq 1$  (c)  $\text{supp}(\phi_n) \uparrow \mathbb{R}^N$  as  $n \rightarrow \infty$ .

By Step 1,

$$M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx))$$

for every  $n \geq 1$ .

By Theorem 5.5,

$$M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} \rightarrow M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$$

in  $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$  as  $n \rightarrow \infty$ . By the usual Cwikel estimate,

$$M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \rightarrow M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}$$

in  $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$  as  $n \rightarrow \infty$ . Thus

$$\begin{aligned} \lim_{n \rightarrow \infty} M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} &= \\ &= M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}, \end{aligned}$$

where the limit is understood in  $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ . Since the separable part  $(\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx))$  is closed in  $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ , the assertion follows.  $\square$

### 9.3. Proof of Theorem 9.1.

**Lemma 9.7.** *We have*

$$[R_{\text{Dunkl},j}, M_f] - [\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f] \in \mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

*Proof.* Set

$$h(x) = 1 - \frac{|x|}{(1 + |x|^2)^{\frac{1}{2}}}, \quad x \in \mathbb{R}^N.$$

We, obviously, have

$$R_{\text{Dunkl},j} - \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} = R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}).$$

Thus,

$$\begin{aligned} [R_{\text{Dunkl},j}, M_f] - [\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f] &= \\ &= R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}) M_f - M_f h(\nabla_{\text{Dunkl}}) \cdot R_{\text{Dunkl},j} = \\ &= R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}) (1 + \Delta_{\text{Dunkl}}) \cdot (1 + \Delta_{\text{Dunkl}})^{-1} M_f - \\ &\quad - M_f (1 + \Delta_{\text{Dunkl}})^{-1} \cdot (1 + \Delta_{\text{Dunkl}}) h(\nabla_{\text{Dunkl}}) \cdot R_{\text{Dunkl},j}. \end{aligned}$$

In the first summand on the right hand side, the first and the second factors are bounded and the third one belongs to  $\mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx))$  by Lemma 8.6. In the second summand on the right hand side, the second and the third factors are bounded and the first one belongs to  $\mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx))$  by Lemma 8.6.  $\square$

**Lemma 9.8.** *Let  $\{B_j\}_{j=1}^N$  and  $\{D_j\}_{j=1}^N$  be commuting tuples of self-adjoint operators. Denote  $\langle B \rangle = (1 + \sum_{j=1}^N B_j^2)^{\frac{1}{2}}$  and  $\langle D \rangle = (1 + \sum_{j=1}^N D_j^2)^{\frac{1}{2}}$ . Suppose that there exists a sequence  $\{\phi_n\}_{n \geq 1} \subset C_c^\infty(\mathbb{R}^N)$  with*

- (i)  $0 \leq \phi_n \leq 1$  for  $n \geq 1$  and  $\text{supp}(\phi_n) \uparrow \mathbb{R}^N$  as  $n \rightarrow \infty$ ;
- (ii)

$$M_{\phi_n} \langle B \rangle^{-1} - M_{\phi_n} \langle D \rangle^{-1} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)), \quad n \geq 1;$$

- (iii)  $M_{\phi_n}(B_k - D_k)$  is bounded<sup>4</sup> and compactly supported for  $n \geq 1$  and  $1 \leq k \leq K$ ;
- (iv)  $[D_k, M_{f\phi_n}]$  is bounded<sup>5</sup> and compactly supported for  $n \geq 1$  and  $1 \leq k \leq K$ ;
- (v)  $[\langle D \rangle^{-1}, M_{f\phi_n}] \in (\mathcal{L}_{N,\infty})_0(\mathbb{R}^N, w(x)dx)$  for every  $n \geq 1$ ;
- (vi)

$$\|M_h \langle B \rangle^{-1}\|_{\mathcal{L}_{N,\infty}(\mathbb{R}^N, w(x)dx)} \leq c_N \|h\|_{L_N(\mathbb{R}^N)}, \quad h \in L_N(\mathbb{R}^N),$$

$$\|M_h \langle D \rangle^{-1}\|_{\mathcal{L}_{N,\infty}(\mathbb{R}^N, w(x)dx)} \leq c_N \|h\|_{L_N(\mathbb{R}^N)}, \quad h \in L_N(\mathbb{R}^N);$$

We have

$$\frac{B_j}{\langle B \rangle} M_f \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_f \frac{D_k}{\langle D \rangle^2} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

*Proof.* Choose real-valued  $\psi_n \in C_c^\infty(\mathbb{R}^N)$  such that

$$M_{\phi_n}(B_k - D_k) = M_{\phi_n}(B_k - D_k) \cdot M_{\psi_n}, \quad 1 \leq k \leq K,$$

$$[M_{f\phi_n}, D_k] = M_{\psi_n} \cdot [M_{f\phi_n}, D_k] \cdot M_{\psi_n}, \quad 1 \leq k \leq K.$$

We have

$$\begin{aligned} & \frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} = \frac{B_j}{\langle B \rangle} \cdot M_f (M_{\phi_n} \langle B \rangle^{-1} - M_{\phi_n} \langle D \rangle^{-1}) \cdot \frac{B_k}{\langle B \rangle} + \\ & + \frac{B_j}{\langle B \rangle} \cdot [M_{f\phi_n}, \langle D \rangle^{-1}] \cdot \frac{B_k}{\langle B \rangle} + \frac{B_j}{\langle B \rangle} \cdot \langle D \rangle^{-1} M_f \cdot M_{\phi_n}(B_k - D_k) \cdot M_{\psi_n} \langle B \rangle^{-1} + \\ & + \frac{B_j}{\langle B \rangle} \cdot \langle D \rangle^{-1} M_{\psi_n} \cdot [M_{f\phi_n}, D_k] \cdot M_{\psi_n} \langle B \rangle^{-1} + \frac{B_j}{\langle B \rangle} \frac{D_k}{\langle D \rangle} \cdot M_f (M_{\phi_n} \langle B \rangle^{-1} - M_{\phi_n} \langle D \rangle^{-1}) + \\ & + \frac{B_j}{\langle B \rangle} \cdot \langle D \rangle^{-1} M_{\psi_n} \cdot [D_k, M_{f\phi_n}] \cdot M_{\psi_n} \langle D \rangle^{-1} + \frac{B_j}{\langle B \rangle} \cdot [\langle D \rangle^{-1}, M_{f\phi_n}] \cdot \frac{D_k}{\langle D \rangle} + \\ & + \langle B \rangle^{-1} M_{\psi_n} \cdot (B_j - D_j) M_{\phi_n} \cdot M_f \frac{D_k}{\langle D \rangle^2} + \\ & + \langle B \rangle^{-1} M_{\psi_n} \cdot [D_j, M_{f\phi_n}] \cdot M_{\psi_n} \frac{D_k}{\langle D \rangle^2} + (\langle B \rangle^{-1} - \langle D \rangle^{-1}) M_{\phi_n} \cdot M_f \frac{D_j D_k}{\langle D \rangle^2} + \\ & + \langle D \rangle^{-1} M_{\psi_n} \cdot [M_{f\phi_n}, D_j] \cdot M_{\psi_n} \frac{D_k}{\langle D \rangle^2}. \end{aligned}$$

It follows now from the assumptions and from Hölder inequality that

$$\frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

Obviously,

$$\left\| \frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq$$

<sup>4</sup>By a constant which depends also on  $n$ .

<sup>5</sup>By a constant which depends also on  $n$ .

$$\begin{aligned}
&\leq \left\| M_{f(1-\phi_n)} \langle B \rangle^{-1} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_N \|f(1-\phi_n)\|_{L_N(\mathbb{R}^N)}, \\
&\quad \left\| \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} - \frac{D_j}{\langle D \rangle} M_f \frac{D_k}{\langle D \rangle^2} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\
&\leq \left\| M_{f(1-\phi_n)} \langle D \rangle^{-1} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_N \|f(1-\phi_n)\|_{L_N(\mathbb{R}^N)}.
\end{aligned}$$

Thus,

$$\begin{aligned}
&\lim_{n \rightarrow \infty} \frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} = \\
&= \frac{B_j}{\langle B \rangle} M_f \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_f \frac{D_k}{\langle D \rangle^2},
\end{aligned}$$

where the limit is take in  $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ . Since  $(\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx))$  is closed in  $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ , the assertion follows.  $\square$

**Lemma 9.9.** For every  $f \in C_c^\infty(\mathbb{R}^N)$ ,

$$\begin{aligned}
&\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} M_f \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} U_v \in \\
&\in M_{w^{-\frac{1}{2}}} R_j M_f R_k U_v \Delta^{-\frac{1}{2}} M_{w^{\frac{1}{2}}} + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)), \\
&\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} M_f U_v \frac{T_{e_j}}{1 + \Delta_{\text{Dunkl}}} \in \\
&\in M_{w^{-\frac{1}{2}}} R_j M_f U_v R_k \Delta^{-\frac{1}{2}} M_{w^{\frac{1}{2}}} + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

*Proof.* The assertion follows by applying Lemma 9.8 to the tuples  $(T_{e_1}, \dots, T_{e_N})$  and  $(M_{w^{-\frac{1}{2}}} \partial_1 M_{w^{\frac{1}{2}}}, \dots, M_{w^{-\frac{1}{2}}} \partial_N M_{w^{\frac{1}{2}}})$  and sequence  $\{\phi_n\}_{n \geq 1} \subset C_c^\infty(\mathbb{R}^N)$  supported away from the hyperplanes  $\{x : \langle x, v \rangle = 0\}_{v \in R}$ .  $\square$

*Proof of Theorem 9.1.* We have

$$\begin{aligned}
&[\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f] = \\
&= [T_{e_j}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot [(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}.
\end{aligned}$$

Using Lemma 9.7 and Theorem 9.2, we write

$$\begin{aligned}
&[R_{\text{Dunkl},j}, M_f] \in [T_{e_j}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \\
&-\frac{1}{2} \cdot \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot [\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

It follows from Lemma 8.2 and Lemma 8.6 that

$$\begin{aligned}
&[\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} - 2 \sum_{k=1}^N M_{\partial_k f} \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} - \\
&-\frac{1}{2} \sum_{k=1}^N \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} \left( \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} U_v + U_v \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} \right) \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

Thus,

$$[R_{\text{Dunkl},j}, M_f] \in M_{\partial_j f} (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + \frac{1}{2} \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} U_v -$$

$$\begin{aligned}
& -\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot \sum_{k=1}^N M_{\partial_k f} \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} - \\
& -\frac{1}{4} \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot \sum_{k=1}^N \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} \left( \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} U_v + U_v \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} \right) \in \\
& \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

The assertion follows from Lemma 9.9.  $\square$

## 10. SPECTRAL ASYMPTOTIC FORMULA

**Theorem 10.1.** *Let  $N > 2$ . Let  $\Pi$  be the  $C^*$ -algebra of operators admitting principal symbol. For every  $T \in \Pi \rtimes_\alpha G$  compactly supported from the right, we have*

$$\lim_{t \rightarrow \infty} t^{\frac{1}{N}} \mu(t, T \Delta^{-\frac{1}{2}}) = c_\alpha \|(\text{sym} \rtimes G)(T)\|_{L_N((L_\infty(\mathbb{R}^N) \otimes L_\infty(\mathbb{S}^{N-1})) \rtimes_\beta \beta)}.$$

here is argument by the chatbot, rewritten by me and checked and re-written by chatbot. so, the question is indeed reduced to matrix-valued FSZ. to be written carefully.

Let  $G$  be a Weyl group, acting orthogonally on  $\mathbb{R}^N$ , and let  $\Gamma$  be a fixed chamber. We use the convention

$$U_g \xi = \xi \circ g^{-1}, \quad g \in G.$$

Thus  $U_g$  is unitary on  $L_2(\mathbb{R}^N)$  and

$$U_g L_2(\Gamma) = L_2(g\Gamma).$$

Let  $\{e_g\}_{g \in G}$  be the standard basis of  $\mathbb{C}^G$ .

Define

$$W : L_2(\mathbb{R}^N) \otimes \mathbb{C}^G \longrightarrow L_2(\mathbb{R}^N)$$

by

$$W \left( \sum_{g \in G} \xi_g \otimes e_g \right) = \sum_{g \in G} U_g M_{\chi_\Gamma} \xi_g = \sum_{g \in G} (\xi_g|_\Gamma) \circ g^{-1}.$$

Since the chambers  $g\Gamma$  partition  $\mathbb{R}^N$  up to a null set, we have

$$WW^* = \text{id}_{L_2(\mathbb{R}^N)}, \quad W^*W = M_{\chi_\Gamma} \otimes \text{id}_{\mathbb{C}^G}.$$

Consequently, for the singular-value distribution  $\mu$ ,

$$\mu(W^*TW) = \mu(T), \quad T \in B(L_2(\mathbb{R}^N)),$$

where the equality is understood after discarding the zero singular values coming from  $\ker W$ .

**Lemma 10.2.** *If  $f \in C_c^\infty(\mathbb{R}^N)$ , then*

$$M_f \Delta^{-1/2} \in \sum_{g \in G} M_{f\chi_{g\Gamma}} \Delta^{-1/2} M_{\chi_{g\Gamma}} + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N)).$$

*Proof.* This follows from Theorem 4.1 in FSZ-TAMS.  $\square$

**Lemma 10.3.** *If  $f \in C_c^\infty(\mathbb{R}^N)$  and  $h \in C^\infty(\mathbb{S}^{N-1})$ , then*

$$h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_f \in \sum_{g \in G} M_{\chi_{g\Gamma}} h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{f\chi_{g\Gamma}} + \mathcal{K}(L_2(\mathbb{R}^N)).$$

*Proof.* This is true. We need to provide a proof or a reference.  $\square$

**Lemma 10.4.** *Let  $f \in C_c^\infty(\mathbb{R}^N)$  and let  $g \in G$ . We have*

$$W^* M_f W = \sum_{g \in G} M_{(f\chi_{g\Gamma}) \circ g} \otimes E_{g,g},$$

$$W^* \left( M_{\chi_{g\Gamma}} \Delta^{-1/2} M_{\chi_{g\Gamma}} \right) W = M_{\chi_\Gamma} \Delta^{-1/2} M_{\chi_\Gamma} \otimes E_{g,g},$$

and

$$W^* M_{\chi_{g\Gamma}} h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{\chi_{g\Gamma}} W = M_{\chi_\Gamma} (h \circ g) \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{\chi_\Gamma} \otimes E_{g,g}.$$

Moreover,

$$W^* U_g W = \sum_{g' \in G} M_{\chi_\Gamma} \otimes E_{gg',g'}.$$

*Proof.* Write

$$P = M_{\chi_\Gamma}.$$

Then

$$W = \sum_{g \in G} U_g P \otimes \langle e_g, \cdot \rangle, \quad W^* = \sum_{g \in G} P U_g^* \otimes |e_g\rangle.$$

For the first identity, the  $(g, g)$  matrix entry is

$$P U_g^* M_f U_g P = P M_{f \circ g} P = M_{\chi_\Gamma (f \circ g)}.$$

Since

$$\chi_\Gamma (f \circ g) = (f \chi_{g\Gamma}) \circ g,$$

we obtain

$$W^* M_f W = \sum_{g \in G} M_{(f\chi_{g\Gamma}) \circ g} \otimes E_{g,g}.$$

Similarly,

$$P U_g^* M_{\chi_{g\Gamma}} U_g P = P,$$

and, since  $\Delta^{-1/2}$  is invariant under orthogonal changes of variables,

$$U_g^* \Delta^{-1/2} U_g = \Delta^{-1/2}.$$

Hence

$$W^* \left( M_{\chi_{g\Gamma}} \Delta^{-1/2} M_{\chi_{g\Gamma}} \right) W = M_{\chi_\Gamma} \Delta^{-1/2} M_{\chi_\Gamma} \otimes E_{g,g}.$$

For the Fourier multiplier, we use

$$U_g^* h \left( \frac{\nabla}{\Delta^{1/2}} \right) U_g = (h \circ g) \left( \frac{\nabla}{\Delta^{1/2}} \right).$$

Therefore

$$W^* M_{\chi_{g\Gamma}} h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{\chi_{g\Gamma}} W = M_{\chi_\Gamma} (h \circ g) \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{\chi_\Gamma} \otimes E_{g,g}.$$

Finally, the  $(k, g')$  matrix entry of  $W^* U_g W$  is

$$P U_k^* U_g U_{g'} P.$$

This vanishes unless

$$k = gg',$$

in which case it equals  $P$ . Hence

$$W^* U_g W = \sum_{g' \in G} M_{\chi_\Gamma} \otimes E_{gg', g'}.$$

□

**Lemma 10.5.** *Modulo the separable part of  $\mathcal{L}_{N, \infty}$ , one has*

$$\begin{aligned} W^* h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_f \Delta^{-1/2} W \\ \equiv \sum_{g \in G} \left[ M_{\chi_\Gamma} (h \circ g) \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{(f\chi_{g\Gamma}) \circ g} \Delta^{-1/2} M_{\chi_\Gamma} \right] \otimes E_{g, g}. \end{aligned}$$

*Proof.* By the second lemma,

$$h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_f \equiv \sum_{g \in G} M_{\chi_{g\Gamma}} h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{f\chi_{g\Gamma}}$$

modulo the separable part. Hence

$$h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_f \Delta^{-1/2} \equiv \sum_{g \in G} M_{\chi_{g\Gamma}} h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{f\chi_{g\Gamma}} \Delta^{-1/2}.$$

Applying the first lemma to the localized multiplier and multiplying on the left by  $M_{\chi_{g\Gamma}}$ , all chamber terms except the  $g$ -th one vanish. Thus

$$M_{\chi_{g\Gamma}} h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{f\chi_{g\Gamma}} \Delta^{-1/2} \equiv M_{\chi_{g\Gamma}} h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{f\chi_{g\Gamma}} \Delta^{-1/2} M_{\chi_{g\Gamma}}.$$

Consequently,

$$h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_f \Delta^{-1/2} \equiv \sum_{g \in G} M_{\chi_{g\Gamma}} h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{f\chi_{g\Gamma}} \Delta^{-1/2} M_{\chi_{g\Gamma}}.$$

Each summand maps  $L_2(g\Gamma)$  into  $L_2(g\Gamma)$ . Therefore its conjugate by  $W$  is diagonal. Its  $g$ -th diagonal entry is

$$M_{\chi_\Gamma} U_g^* h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{f\chi_{g\Gamma}} \Delta^{-1/2} U_g M_{\chi_\Gamma}.$$

Using

$$U_g^* h \left( \frac{\nabla}{\Delta^{1/2}} \right) U_g = (h \circ g) \left( \frac{\nabla}{\Delta^{1/2}} \right)$$

and

$$U_g^* M_{f\chi_{g\Gamma}} U_g = M_{(f\chi_{g\Gamma}) \circ g},$$

we obtain

$$M_{\chi_\Gamma} (h \circ g) \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{(f\chi_{g\Gamma}) \circ g} \Delta^{-1/2} M_{\chi_\Gamma}.$$

This proves the claim. □

**Lemma 10.6.** *Modulo the separable part of  $\mathcal{L}_{N, \infty}$ , one has*

$$W^* h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_f \Delta^{-1/2} U_g W =$$

$$= \sum_{g' \in G} \left[ (h \circ g') \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{(f\chi_{g'\Gamma}) \circ g'} \Delta^{-1/2} \right] \otimes E_{g', g^{-1}g'}.$$

*Proof.*

$$\begin{aligned} & W^* h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_f \Delta^{-1/2} W \\ & \equiv \sum_{g' \in G} \left[ M_{\chi_\Gamma} (h \circ g') \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{(f\chi_{g'\Gamma}) \circ g'} \Delta^{-1/2} M_{\chi_\Gamma} \right] \otimes E_{g', g'}. \end{aligned}$$

On the other hand, we have

$$W^* U_g W = \sum_{g'' \in G} M_{\chi_\Gamma} \otimes E_{gg'', g''}.$$

Thus,

$$\begin{aligned} & W^* h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_f \Delta^{-1/2} W = \\ & = \sum_{g' \in G} \left[ M_{\chi_\Gamma} (h \circ g') \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{(f\chi_{g'\Gamma}) \circ g'} \Delta^{-1/2} M_{\chi_\Gamma} \right] \otimes E_{g', g'} \times \\ & \quad \times \sum_{g'' \in G} M_{\chi_\Gamma} \otimes E_{gg'', g''}. \end{aligned}$$

We obviously have  $gg'' = g'$  and, hence,  $g'' = g^{-1}g'$ . Hence,

$$\begin{aligned} & W^* h \left( \frac{\nabla}{\Delta^{1/2}} \right) M_f \Delta^{-1/2} W = \\ & = \sum_{g' \in G} \left[ M_{\chi_\Gamma} (h \circ g') \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{(f\chi_{g'\Gamma}) \circ g'} \Delta^{-1/2} M_{\chi_\Gamma} \right] \otimes E_{g', g^{-1}g'}. \end{aligned}$$

It remains to note that

$$M_{\chi_\Gamma} (h \circ g') \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{(f\chi_{g'\Gamma}) \circ g'} \Delta^{-1/2} M_{\chi_\Gamma} = (h \circ g') \left( \frac{\nabla}{\Delta^{1/2}} \right) M_{(f\chi_{g'\Gamma}) \circ g'} \Delta^{-1/2}$$

modulo separable part.  $\square$

#### APPENDIX A. ZHIJIE LEMMA

**Lemma A.1.** *Let  $1 < s < \infty$ , and let  $\mathbb{D}^0$  be the standard dyadic grid in  $\mathbb{R}^N$ . Suppose that  $F \in L^s(\mathbb{R}^N)$ . Then there exists a constant  $C > 0$  such that*

$$\left\| \left\{ |Q|^{-1/s'} \int_Q |F(x)| dx \right\}_{Q \in \mathbb{D}^0} \right\|_{\ell^{s, \infty}} \leq C \|F\|_{L^s(\mathbb{R}^N)}.$$

*Proof.* Let  $\mathcal{E} \subset \mathbb{D}^0$  be finite and set

$$G_{\mathcal{E}}(x) := \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Since the dyadic cubes containing any fixed  $x$  are totally ordered by inclusion, for each  $x \in \bigcup_{Q \in \mathcal{E}} Q$  we may choose a cube  $Q_x \in \mathcal{E}$  of minimal volume such that

$x \in Q_x$ . Every  $Q \in \mathcal{E}$  containing  $x$  is then a dyadic ancestor of  $Q_x$ . Since the volume increases by the factor  $2^N$  at each generation,

$$G_{\mathcal{E}}(x) = \sum_{\substack{Q \in \mathcal{E} \\ x \in Q}} |Q|^{-1/s'} \leq |Q_x|^{-1/s'} \sum_{k=0}^{\infty} 2^{-kN/s'} \lesssim_{N,s} \sup_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Consequently,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)}^{s'} \lesssim_{N,s} \int_{\mathbb{R}^N} \sup_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx \leq \int_{\mathbb{R}^N} \sum_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx = \#\mathcal{E}.$$

and hence,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)} \lesssim_{N,s} (\#\mathcal{E})^{1/s'}.$$

By Hölder's inequality,

$$\sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx = \int_{\mathbb{R}^N} |F(x)| G_{\mathcal{E}}(x) dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'}.$$

Fix  $t > 0$  and let  $\mathcal{E}$  be any finite subcollection of the dyadic cubes satisfying

$$|Q|^{-1/s'} \int_Q |F(x)| dx > t.$$

Then

$$t \#\mathcal{E} < \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'},$$

so

$$\#\mathcal{E} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s.$$

Since  $\mathcal{E}$  is arbitrary,

$$\#\left\{Q \in \mathbb{D}^0 : |Q|^{-1/s'} \int_Q |F(x)| dx > t\right\} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s, \quad t > 0.$$

The asserted weak  $\ell^s$  estimate follows. This completes the proof of Lemma A.1.  $\square$

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